

```

root_path = fileparts(mfilename("fullpath"));
data_path = fullfile(root_path, 'sym4D_cutsAndFits');

if ~exist('cuts2fit400', 'var')
    ld = load(fullfile(data_path, 'multicuts_fit_dataGH_ei400meV.mat'));
    cuts2fit400 = ld.cuts2fit400;
end
used_ds = containers.Map()

```

```
used_ds =
```

```
Map with properties:
```

```

    Count: 0
    KeyType: char
    ValueType: any

```

```
Show all accessible properties of Map
```

```

sel_info = struct('cuts', [], 'en_range', [], 'fname', '', 'bg_par', []);
%
sel_info.cuts = cuts2fit400.cuts_dir001FFfit(1);
sel_info.en_range = 50:10:160;
sel_info.fname='EnFit_Ei400ref001off110_2D_constFFCut';
sel_info.bg_par= {};
used_ds('Ei400Br110dir001')=sel_info;
%
sel_info.cuts = cuts2fit400.cuts_dir001FFfit(2);
sel_info.en_range = 50:10:200;
sel_info.fname='EnFit_Ei400ref001off200_2D_constFFCut';
sel_info.bg_par= {};
used_ds('Ei400Br200dir001')=sel_info;

```

```

%cuts_list = cuts2fit400.cutsS_list; % all available cuts
%cuts_list = {cuts2fit400.cut_b200_200sel}; % cuts in low ff direction around
<200>
%cuts_list = cuts2fit400.cuts_dir001FFfit(1);
select= 'Ei400Br200dir001';
sel_info = used_ds(select);

cuts_list = sel_info.cuts;

mi = maps_instrument(400,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0], 'cuboid',[0.04,0.03,0.02]);
for i=1:numel(cuts_list)
    cuts_list{i} = cuts_list{i}.set_instrument(mi);
    cuts_list{i} = cuts_list{i}.set_sample(sample);
end

dir_name = "GH";

```

```

dE_step = 4; %original energy transfer step data were binned to. No point in
going finer
half_dE = 10; % half width of data binning
%cut_en = 50:10:260;
%cut_en = 50:10:160; %cuts2fit400.cuts_dir001FFfit(1);
cut_en = sel_info.en_range; %cuts2fit400.cuts_dir001FFfit(2);

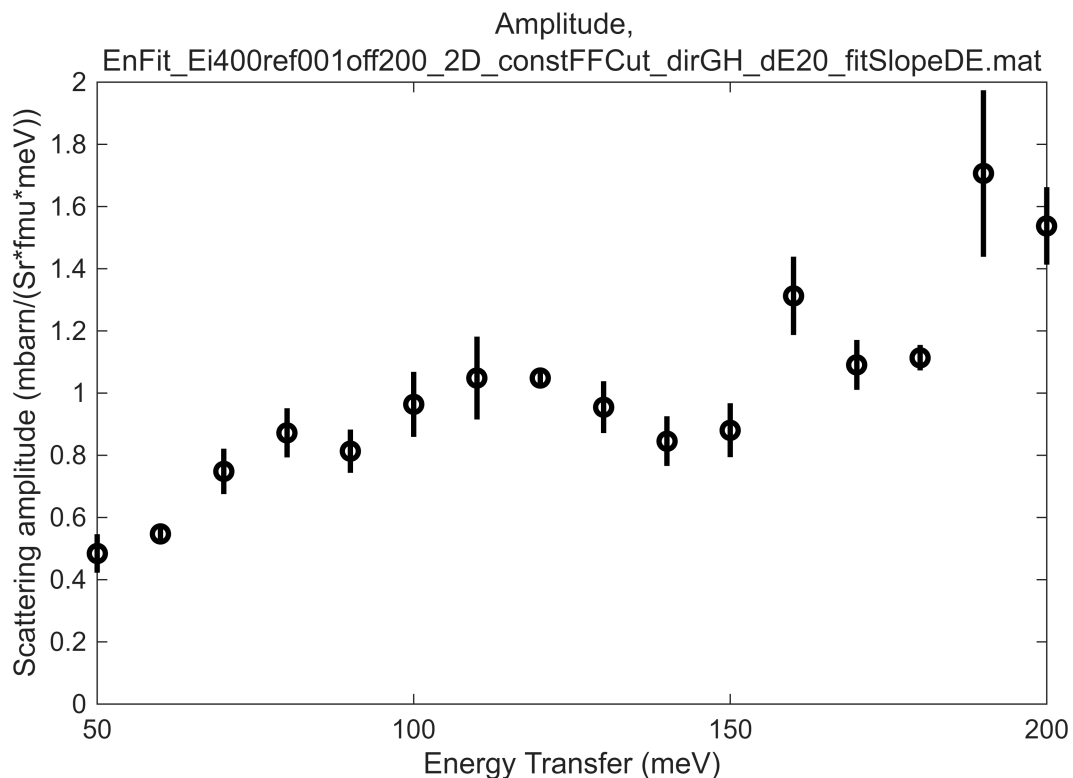
%c1= {cuts_list{3}};
%c1 = {w2_test_cut};
% if isfield(cuts2fit400,'bg_par_010off000')
%     if numel(cuts_list) == 3
%         bg_par =
{cuts2fit400.bg_par_010off000.p,cuts2fit400.bg_par_010off100.p,cuts2fit400.bg_par
_010off200.p};
%     else
%         bg_par = {cuts2fit400.bg_par_010off100.p};
%     end
%     cuts_list{end+1} = bg_par;
% end
bg_par = sel_info.bg_par;
cuts_list{end+1} = bg_par;
fbase_name = sel_info.fname;

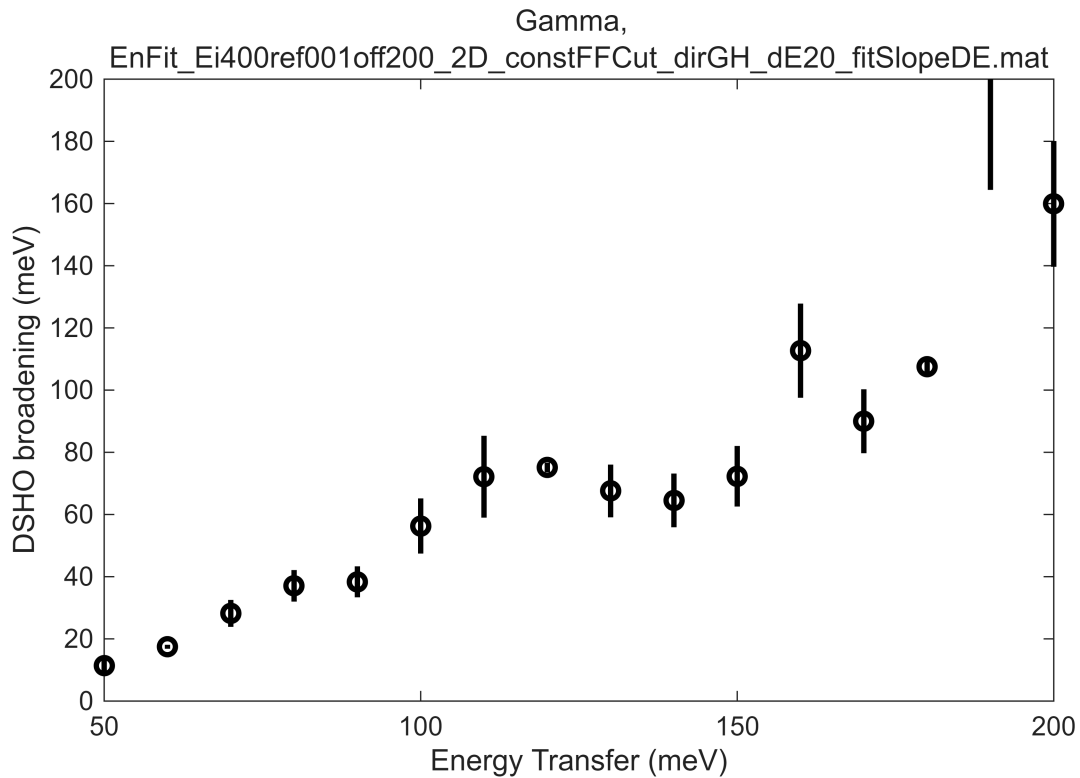
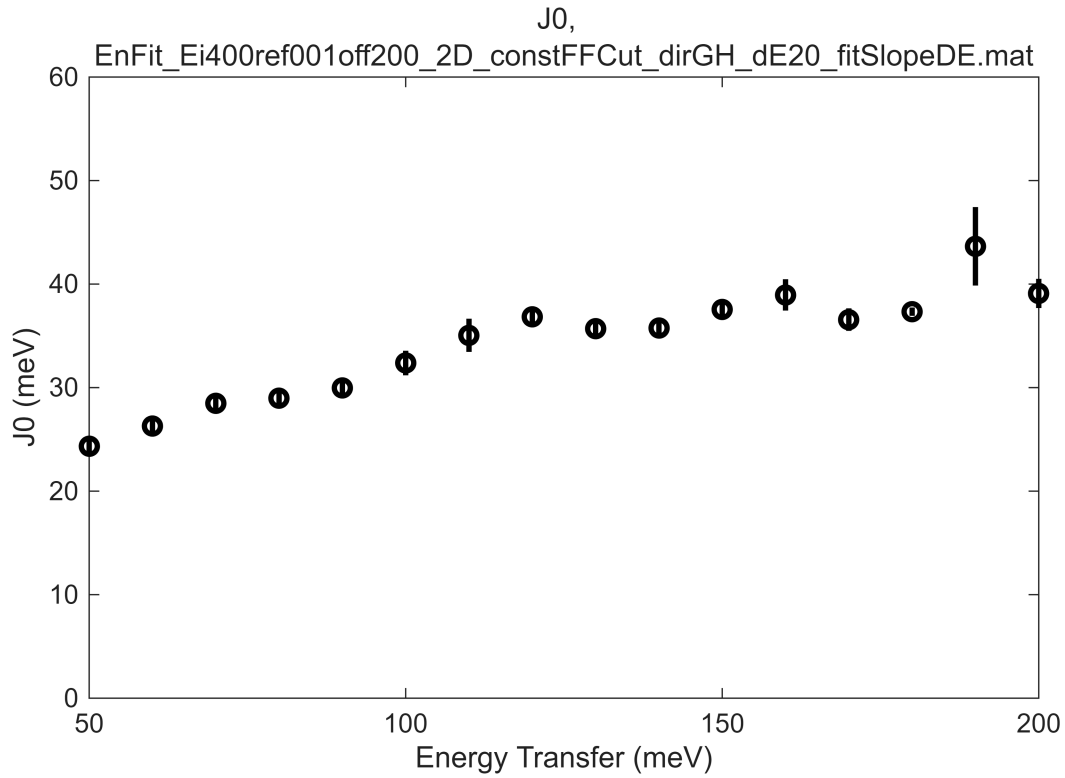
```

```

[fit_res_400rec,res_name] = fit_multicuts_along_direction(...
    cuts_list,fbase_name,dir_name,cut_en,dE_step,half_dE);

```





**** En = 50±10

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 92597 pixels -- processing data... -----> included 75568 pixels

Number of points in input object: 92597

Fraction of object selected: 100.00% (= 92597 points) ; Access speed: 1.85GB/sec

Fraction of object retained: 81.61% (= 75568 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 20.8%; Transf. time frac.: 67.7%; Resulting pix preparation: 0.0%

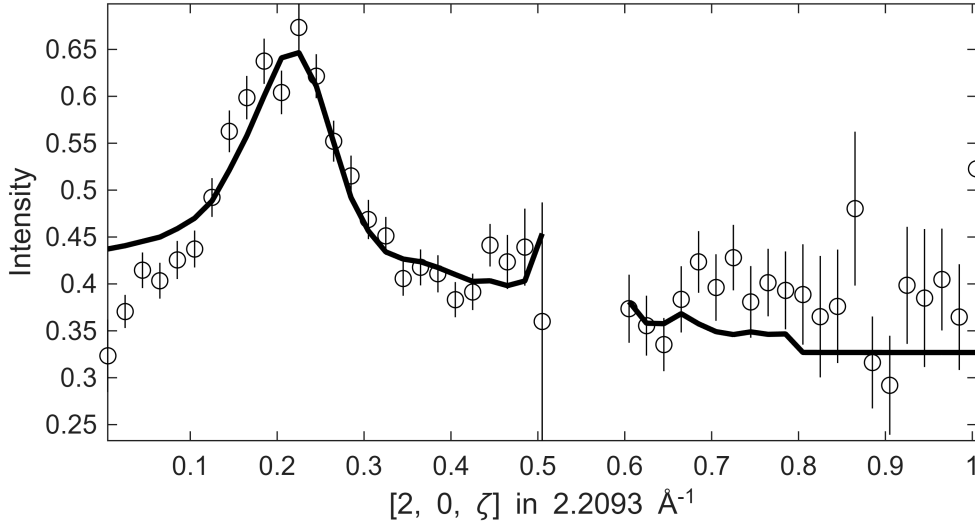
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
 *** Step 1 of 1; Read data for 92597 pixels -- processing data... -----> retained 75568 pixels

Number of points in input object: 92597

Fraction of object selected: 100.00% (= 92597 points) ; Access speed: 1.77GB/sec
 Fraction of object retained: 81.61% (= 75568 points)

**** Total time in cut(sqw) : 0.0sec Including:
 Data access time fraction: 11.6%; Transf. time frac.: 59.0%; Resulting pix preparation: 26.2%

Fe_ei401_align_sym3.sqw
 $S=0.48\pm0.021$; $J_0=24\pm0.51$; $\gamma=11.2658\pm0.448904$
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $40 \leq E \leq 60$
 $\zeta=-0.005:0.02:1.015$ in $[2, 0, \zeta]$



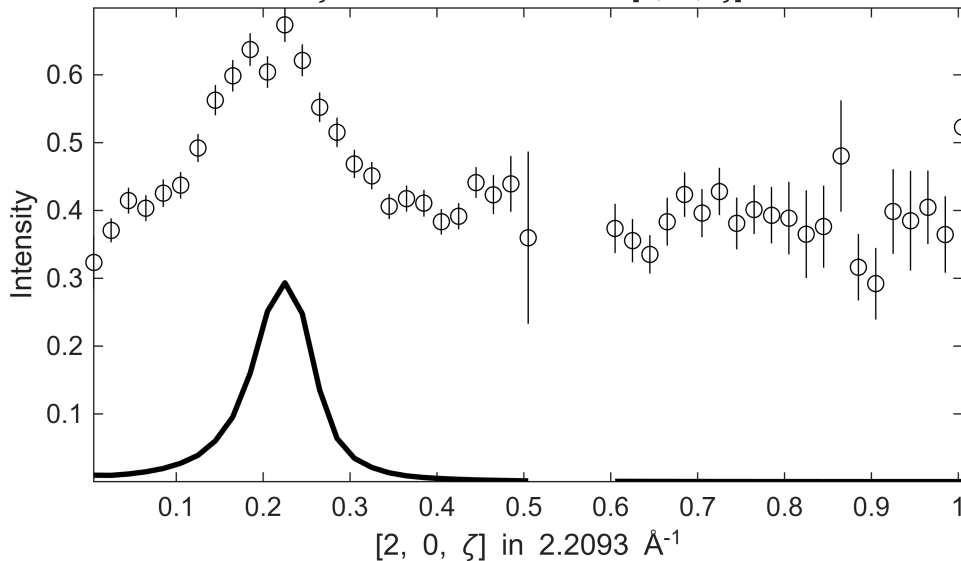
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
 *** Step 1 of 1; Read data for 92597 pixels -- processing data... -----> retained 75568 pixels

Number of points in input object: 92597

Fraction of object selected: 100.00% (= 92597 points) ; Access speed: 2.03GB/sec
 Fraction of object retained: 81.61% (= 75568 points)

**** Total time in cut(sqw) : 0.0sec Including:
 Data access time fraction: 16.9%; Transf. time frac.: 66.1%; Resulting pix preparation: 16.9%

Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $40 \leq E \leq 60$
 $\zeta=-0.005:0.02:1.015$ in $[2, 0, \zeta]$



**** En = 60±10

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 119549 pixels -- processing data... -----> included 101058 pixels

Number of points in input object: 119549

Fraction of object selected: 100.00% (= 119549 points) ; Access speed: 1.93GB/sec

Fraction of object retained: 84.53% (= 101058 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 36.4%; Transf. time frac.: 57.3%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 119549 pixels -- processing data... -----> retained 101058 pixels

Number of points in input object: 119549

Fraction of object selected: 100.00% (= 119549 points) ; Access speed: 1.89GB/sec

Fraction of object retained: 84.53% (= 101058 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 17.4%; Transf. time frac.: 62.8%; Resulting pix preparation: 19.7%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 119549 pixels -- processing data... -----> retained 101058 pixels

Number of points in input object: 119549

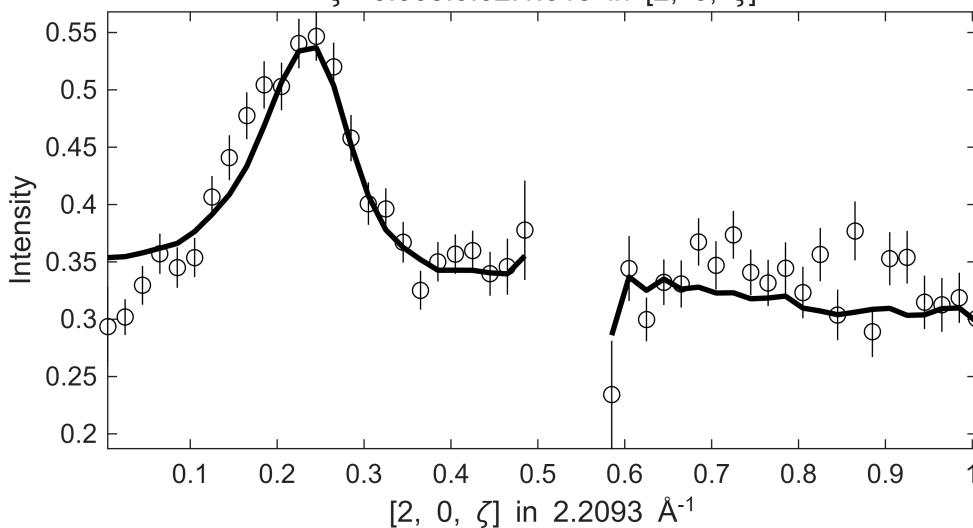
Fraction of object selected: 100.00% (= 119549 points) ; Access speed: 1.79GB/sec

Fraction of object retained: 84.53% (= 101058 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 19.5%; Transf. time frac.: 59.5%; Resulting pix preparation: 20.9%

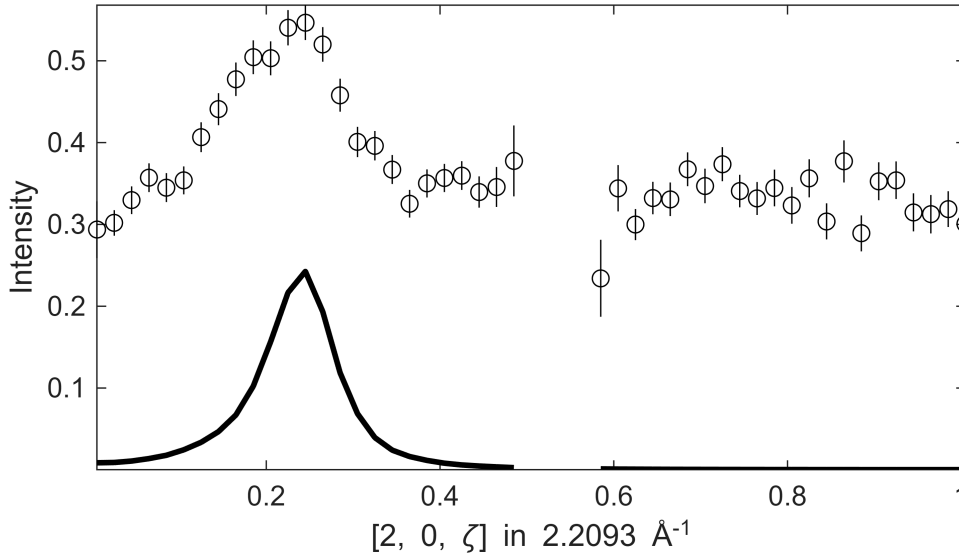
Fe_ei401_align_sym3.sqw
S=0.55±0.021; J0= 26±0.54; gamma=17.5662±0.623532
-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 50 ≤ E ≤ 70
ζ=-0.005:0.02:1.015 in [2, 0, ζ]



Fe_ei401_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $50 \leq E \leq 70$

$\zeta = -0.005:0.02:1.015$ in $[2, 0, \zeta]$



**** En = 70±10

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.3268

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 135496 pixels -- processing data... -----> included 113676 pixels

Number of points in input object: 135496

Fraction of object selected: 100.00% (= 135496 points) ; Access speed: 1.59GB/sec

Fraction of object retained: 83.90% (= 113676 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 36.5%; Transf. time frac.: 58.5%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 135493 pixels -- processing data... -----> retained 113673 pixels

Number of points in input object: 135493

Fraction of object selected: 100.00% (= 135493 points) ; Access speed: 1.79GB/sec

Fraction of object retained: 83.90% (= 113673 points)

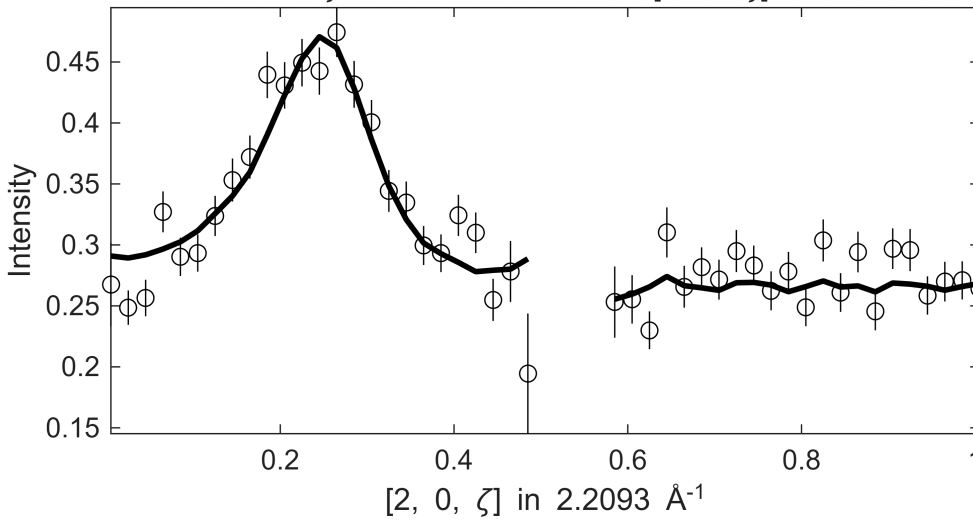
**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 17.4%; Transf. time frac.: 55.2%; Resulting pix preparation: 27.3%

Fe_ei401_align_sym3.sqw

S=0.75±0.073; J0= 28±0.82; gamma=28.3083±4.34847

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 60 ≤ E ≤ 80
ζ=-0.005:0.02:1.015 in [2, 0, ζ]

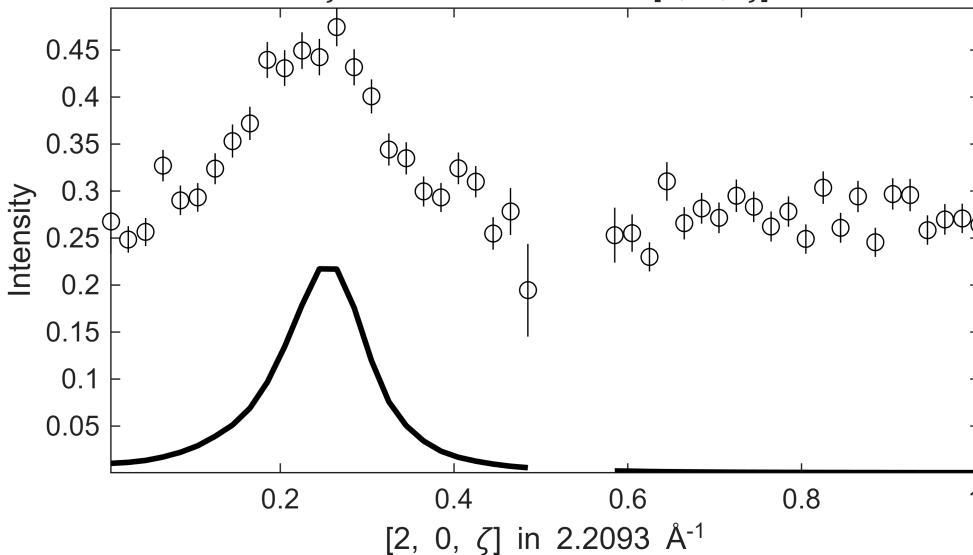


```
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 135496 pixels -- processing data... -----> retained 113676 pixels
-----
Number of points in input object: 135496
  Fraction of object selected: 100.00% (= 135496 points) ; Access speed: 1.45GB/sec
  Fraction of object retained: 83.90% (= 113676 points)

**** Total time in cut(sqw) : 0.0sec Including:
  Data access time fraction: 22.7%; Transf. time frac.: 57.8%; Resulting pix preparation: 19.4%
-----
```

Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 60 ≤ E ≤ 80
ζ=-0.005:0.02:1.015 in [2, 0, ζ]



```
*****
**** En = 80±10
*****
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 134243 pixels -- processing data... -----> included 111170 pixels
-----
Number of points in input object: 134243
  Fraction of object selected: 100.00% (= 134243 points) ; Access speed: 1.89GB/sec
  Fraction of object retained: 82.81% (= 111170 points)

**** Total time in cut(sqw) : 0.0sec Including:
```

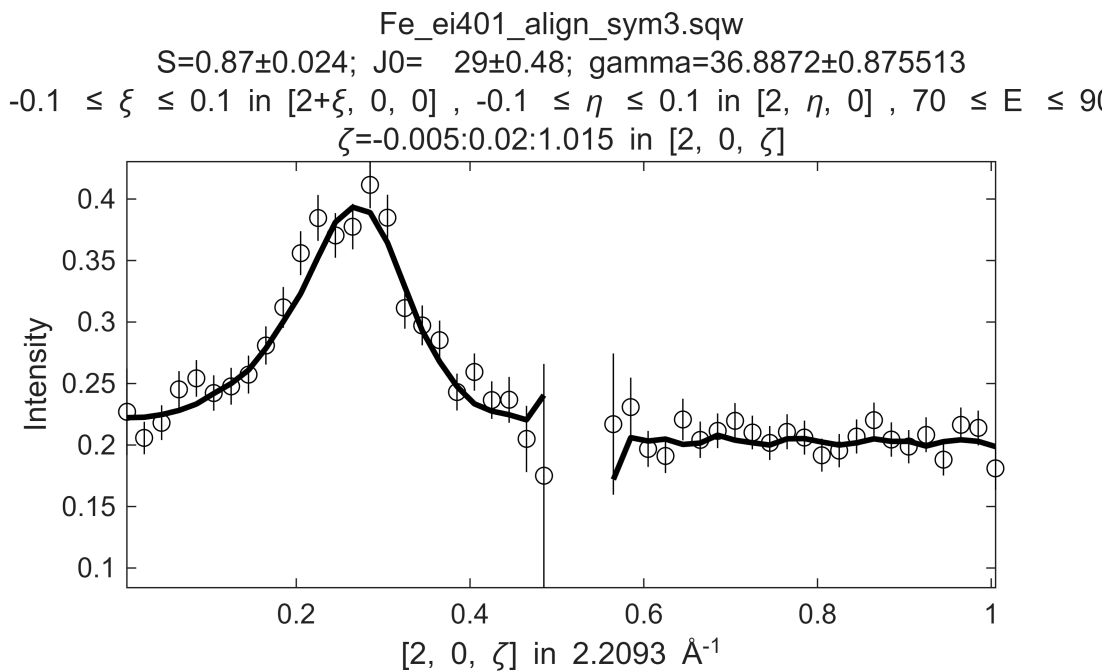
```

Data access time fraction: 38.3%; Transf. time frac.: 55.8%; Resulting pix preparation: 0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 134243 pixels -- processing data... -----> retained 111170 pixels
-----
Number of points in input object: 134243
Fraction of object selected: 100.00% (= 134243 points) ; Access speed: 1.77GB/sec
Fraction of object retained: 82.81% (= 111170 points)

**** Total time in cut(sqw) : 0.0sec Including:
Data access time fraction: 22.7%; Transf. time frac.: 57.3%; Resulting pix preparation: 20.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 134243 pixels -- processing data... -----> retained 111170 pixels
-----
Number of points in input object: 134243
Fraction of object selected: 100.00% (= 134243 points) ; Access speed: 1.54GB/sec
Fraction of object retained: 82.81% (= 111170 points)

**** Total time in cut(sqw) : 0.0sec Including:
Data access time fraction: 23.1%; Transf. time frac.: 55.9%; Resulting pix preparation: 21.0%
-----

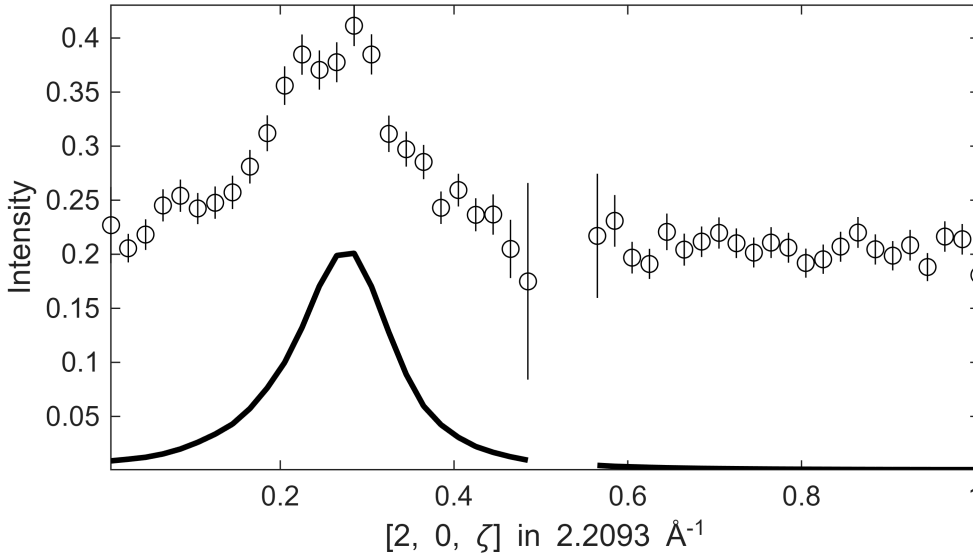
```



Fe_ei401_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $70 \leq E \leq 90$

$\zeta = -0.005:0.02:1.015$ in $[2, 0, \zeta]$



**** En = 90±10

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.3268

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 135827 pixels -- processing data... -----> included 113302 pixels

Number of points in input object: 135827

Fraction of object selected: 100.00% (= 135827 points) ; Access speed: 1.50GB/sec

Fraction of object retained: 83.42% (= 113302 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 36.5%; Transf. time frac.: 58.4%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 135824 pixels -- processing data... -----> retained 113302 pixels

Number of points in input object: 135824

Fraction of object selected: 100.00% (= 135824 points) ; Access speed: 1.48GB/sec

Fraction of object retained: 83.42% (= 113302 points)

**** Total time in cut(sqw) : 0.0sec Including:

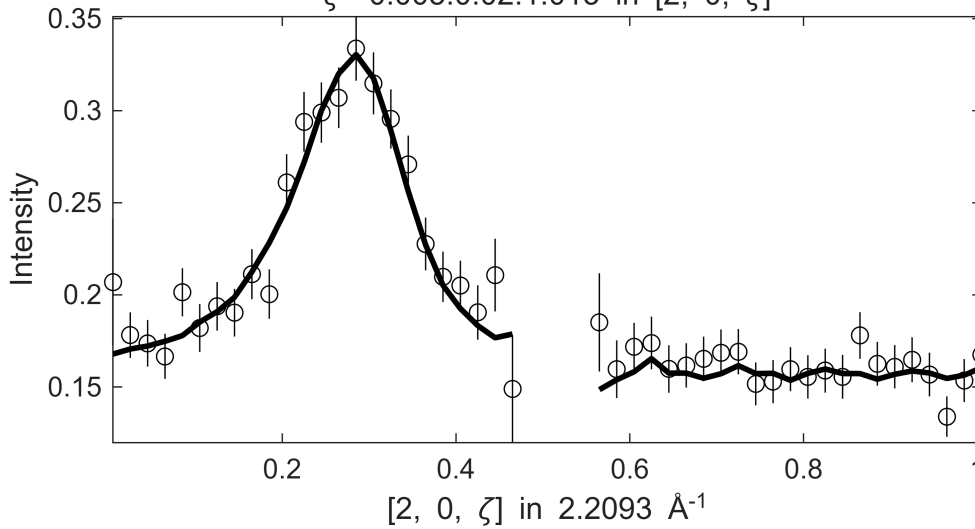
Data access time fraction: 23.8%; Transf. time frac.: 55.7%; Resulting pix preparation: 20.5%

Fe_ei401_align_sym3.sqw

S=0.82±0.07; J0= 30±0.77; gamma=38.4813±4.98873

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 80 ≤ E ≤ 10

ζ=-0.005:0.02:1.015 in [2, 0, ζ]



*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
 *** Step 1 of 1; Read data for 135827 pixels -- processing data... -----> retained 113302 pixels

Number of points in input object: 135827

Fraction of object selected: 100.00% (= 135827 points) ; Access speed: 1.84GB/sec

Fraction of object retained: 83.42% (= 113302 points)

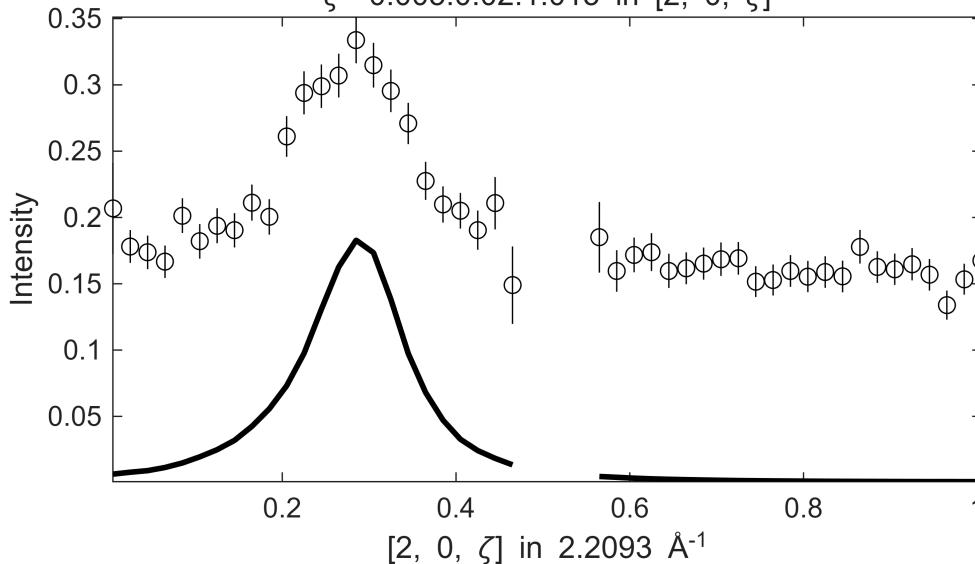
**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 23.2%; Transf. time frac.: 56.3%; Resulting pix preparation: 20.5%

Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 80 ≤ E ≤ 10

ζ=-0.005:0.02:1.015 in [2, 0, ζ]



**** En = 100±10

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.3268

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 142980 pixels -- processing data... -----> included 118727 pixels

Number of points in input object: 142980

Fraction of object selected: 100.00% (= 142980 points) ; Access speed: 1.63GB/sec

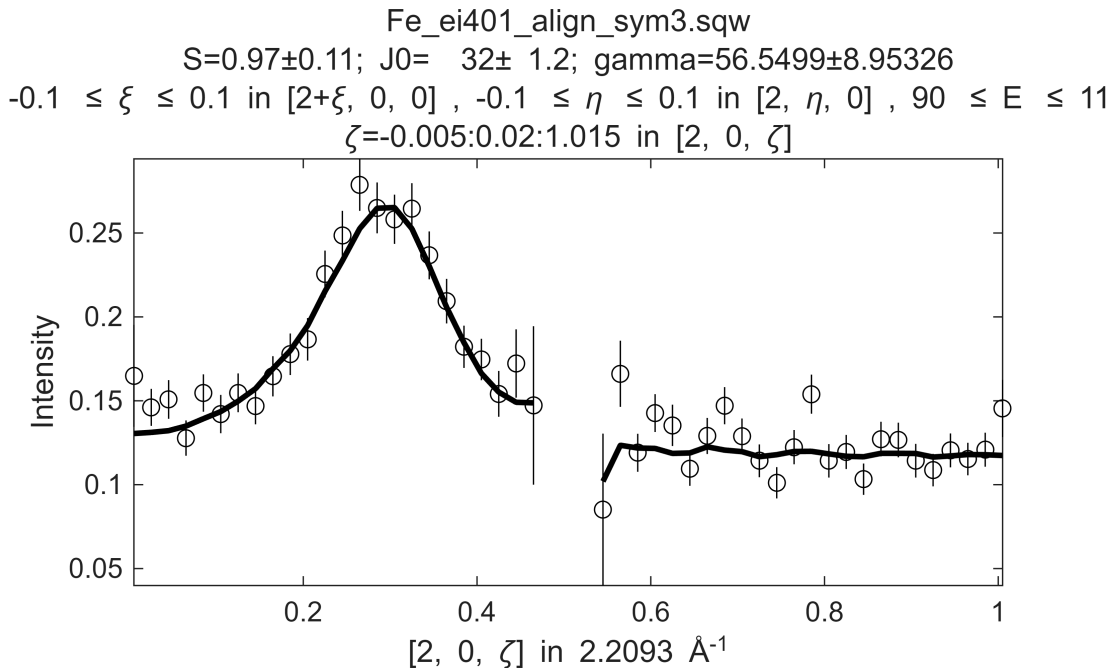
Fraction of object retained: 83.04% (= 118727 points)

```

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 43.4%; Transf. time frac.: 52.2%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 142953 pixels -- processing data... -----> retained 118700 pixels
-----
Number of points in input object: 142953
      Fraction of object selected: 100.00% (= 142953 points) ; Access speed: 1.82GB/sec
      Fraction of object retained:  83.03% (= 118700 points)

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 15.0%; Transf. time frac.: 49.0%; Resulting pix preparation: 35.9%
-----

```



```

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 142980 pixels -- processing data... -----> retained 118727 pixels
-----
Number of points in input object: 142980
      Fraction of object selected: 100.00% (= 142980 points) ; Access speed: 1.59GB/sec
      Fraction of object retained:  83.04% (= 118727 points)

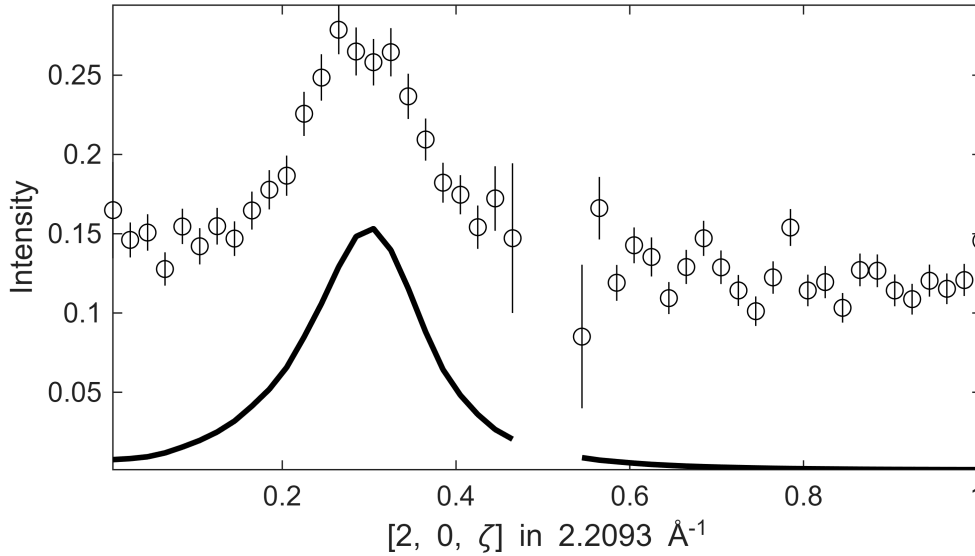
**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 26.5%; Transf. time frac.: 56.2%; Resulting pix preparation: 17.2%
-----

```

Fe_ei401_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $90 \leq E \leq 11$

$\zeta = -0.005:0.02:1.015$ in $[2, 0, \zeta]$



**** En = 110±10

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.3268

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 151100 pixels -- processing data... -----> included 126015 pixels

Number of points in input object: 151100

Fraction of object selected: 100.00% (= 151100 points) ; Access speed: 1.56GB/sec

Fraction of object retained: 83.40% (= 126015 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 34.2%; Transf. time frac.: 62.2%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 151088 pixels -- processing data... -----> retained 126003 pixels

Number of points in input object: 151088

Fraction of object selected: 100.00% (= 151088 points) ; Access speed: 1.52GB/sec

Fraction of object retained: 83.40% (= 126003 points)

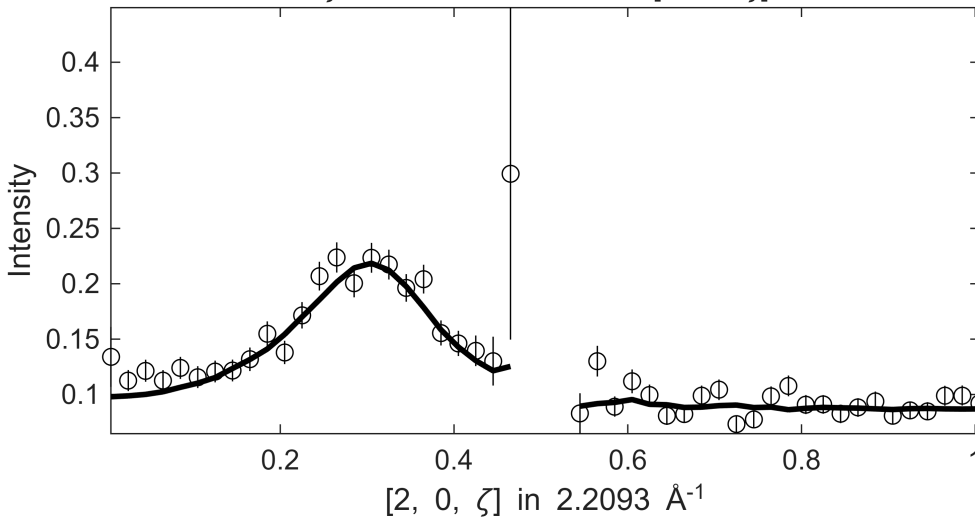
**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 23.6%; Transf. time frac.: 57.3%; Resulting pix preparation: 19.1%

Fe_ei401_align_sym3.sqw

S= 1±0.13; J0= 35± 1.6; gamma=71.932±13.1033

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 100 ≤ E ≤ 1;
ζ=-0.005:0.02:1.015 in [2, 0, ζ]

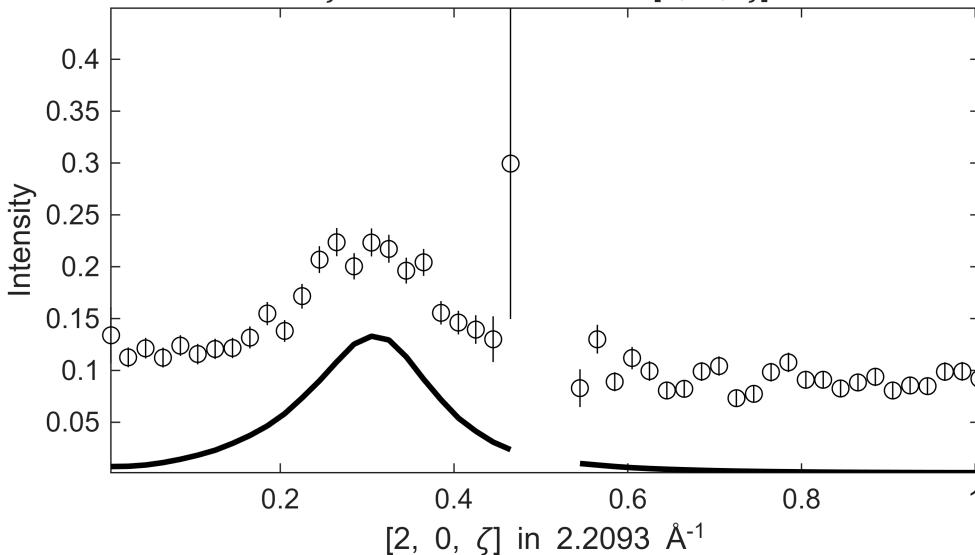


```
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 151100 pixels -- processing data... -----> retained 126015 pixels
-----
Number of points in input object: 151100
Fraction of object selected: 100.00% (= 151100 points) ; Access speed: 1.50GB/sec
Fraction of object retained: 83.40% (= 126015 points)

**** Total time in cut(sqw) : 0.0sec Including:
Data access time fraction: 20.2%; Transf. time frac.: 56.7%; Resulting pix preparation: 23.0%
-----
```

Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 100 ≤ E ≤ 1;
ζ=-0.005:0.02:1.015 in [2, 0, ζ]



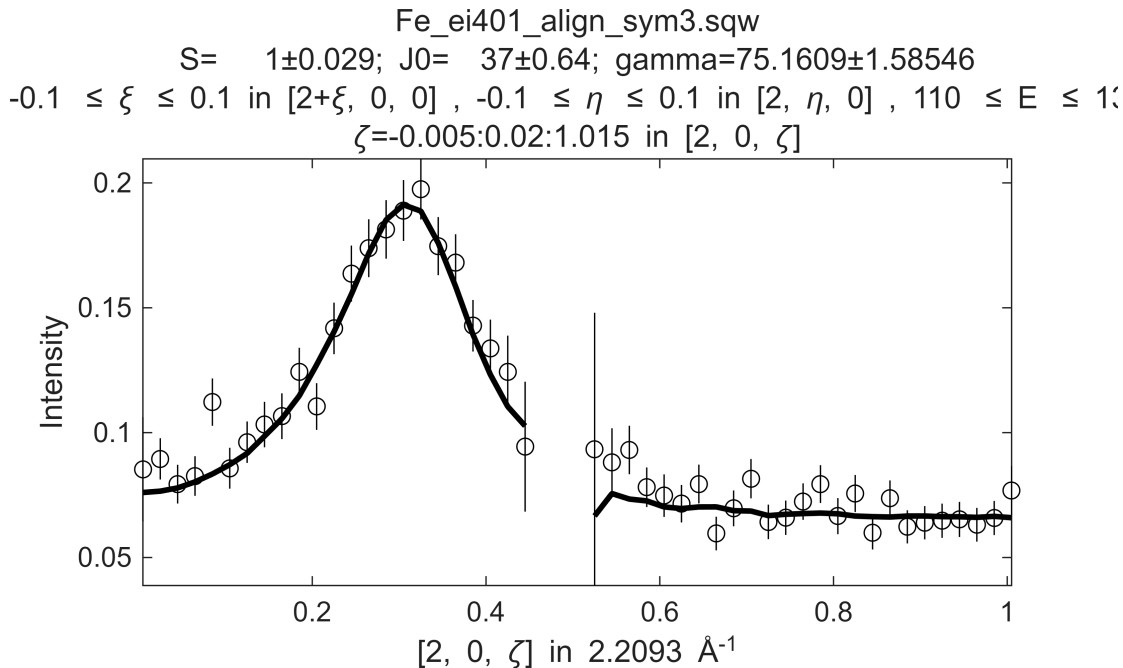
```
*****
**** En = 120±10
*****
Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 0.65359
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 165214 pixels -- processing data... -----> included 137175 pixels
-----
Number of points in input object: 165214
Fraction of object selected: 100.00% (= 165214 points) ; Access speed: 1.99GB/sec
Fraction of object retained: 83.03% (= 137175 points)
```

```

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 39.6%; Transf. time frac.: 55.6%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 165194 pixels -- processing data... -----> retained 137158 pixels
-----
Number of points in input object: 165194
      Fraction of object selected: 100.00% (= 165194 points) ; Access speed: 1.86GB/sec
      Fraction of object retained:  83.03% (= 137158 points)

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 23.5%; Transf. time frac.: 56.1%; Resulting pix preparation: 20.4%
-----

```



```

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 165214 pixels -- processing data... -----> retained 137175 pixels
-----
Number of points in input object: 165214
      Fraction of object selected: 100.00% (= 165214 points) ; Access speed: 1.55GB/sec
      Fraction of object retained:  83.03% (= 137175 points)

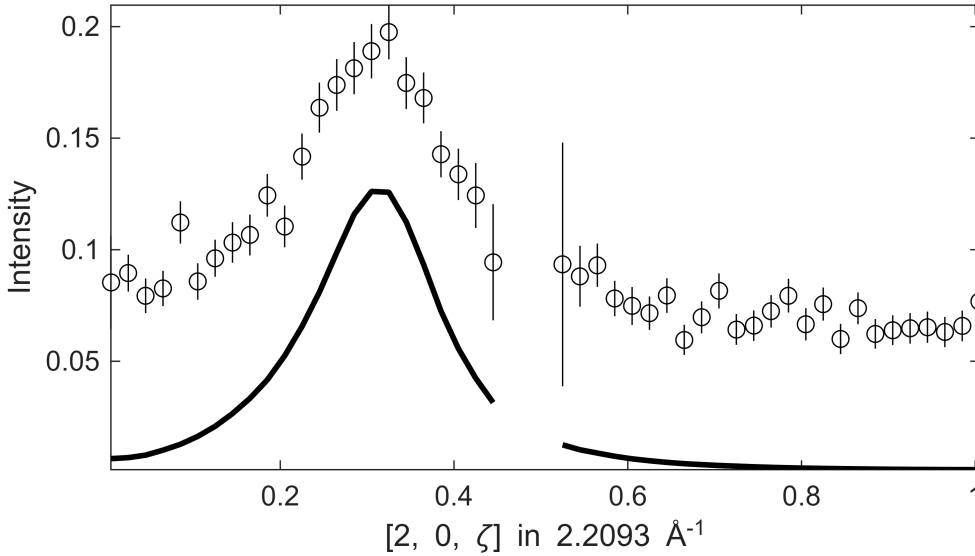
**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 24.1%; Transf. time frac.: 55.9%; Resulting pix preparation: 20.0%
-----

```

Fe_ei401_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $110 \leq E \leq 115$

$\zeta = -0.005:0.02:1.015$ in $[2, 0, \zeta]$



**** En = 130±10

Warning: Dataset:3 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.98039

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 185438 pixels -- processing data... -----> included 154602 pixels

Number of points in input object: 185438

Fraction of object selected: 100.00% (= 185438 points) ; Access speed: 2.00GB/sec

Fraction of object retained: 83.37% (= 154602 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 37.9%; Transf. time frac.: 58.2%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 185192 pixels -- processing data... -----> retained 154356 pixels

Number of points in input object: 185192

Fraction of object selected: 100.00% (= 185192 points) ; Access speed: 1.97GB/sec

Fraction of object retained: 83.35% (= 154356 points)

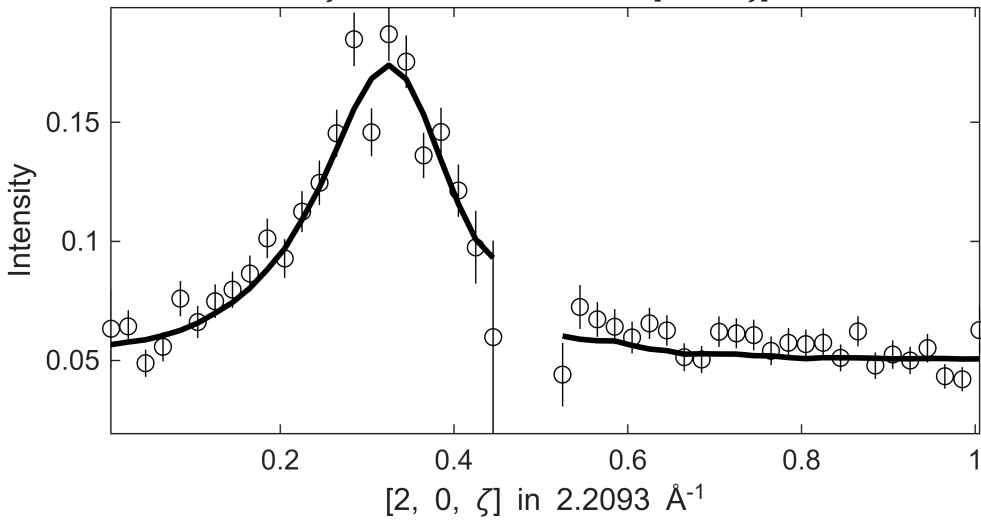
**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 25.0%; Transf. time frac.: 55.3%; Resulting pix preparation: 19.6%

Fe_ei401_align_sym3.sqw

S=0.96±0.085; J0= 36±0.91; gamma=68.2102±8.60968

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 120 ≤ E ≤ 1400 eV
 ζ=-0.005:0.02:1.015 in [2, 0, ζ]

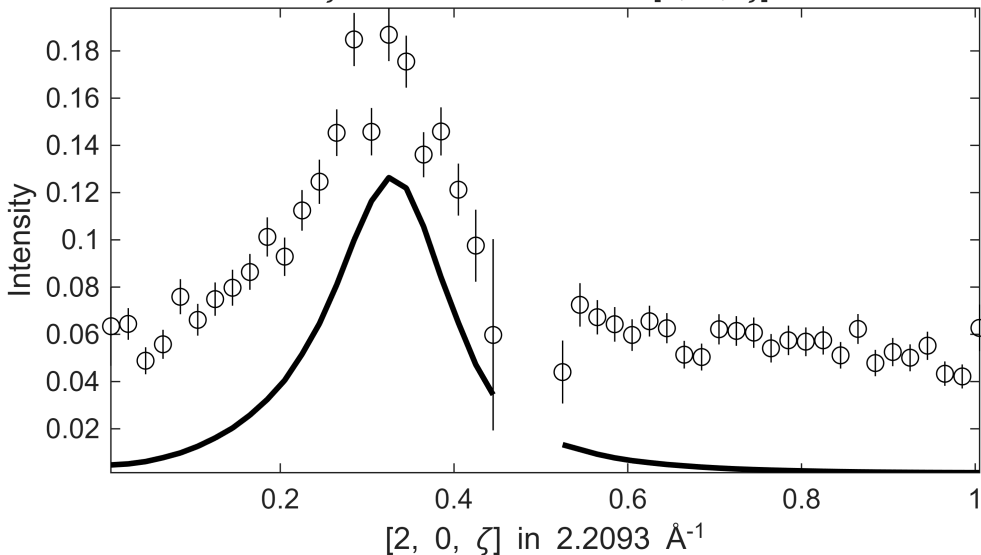


```
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 185438 pixels -- processing data... -----> retained 154602 pixels
-----
Number of points in input object: 185438
  Fraction of object selected: 100.00% (= 185438 points) ; Access speed: 1.61GB/sec
  Fraction of object retained: 83.37% (= 154602 points)

**** Total time in cut(sqw) : 0.0sec Including:
  Data access time fraction: 29.4%; Transf. time frac.: 50.3%; Resulting pix preparation: 20.2%
-----
```

Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 120 ≤ E ≤ 1400 eV
 ζ=-0.005:0.02:1.015 in [2, 0, ζ]



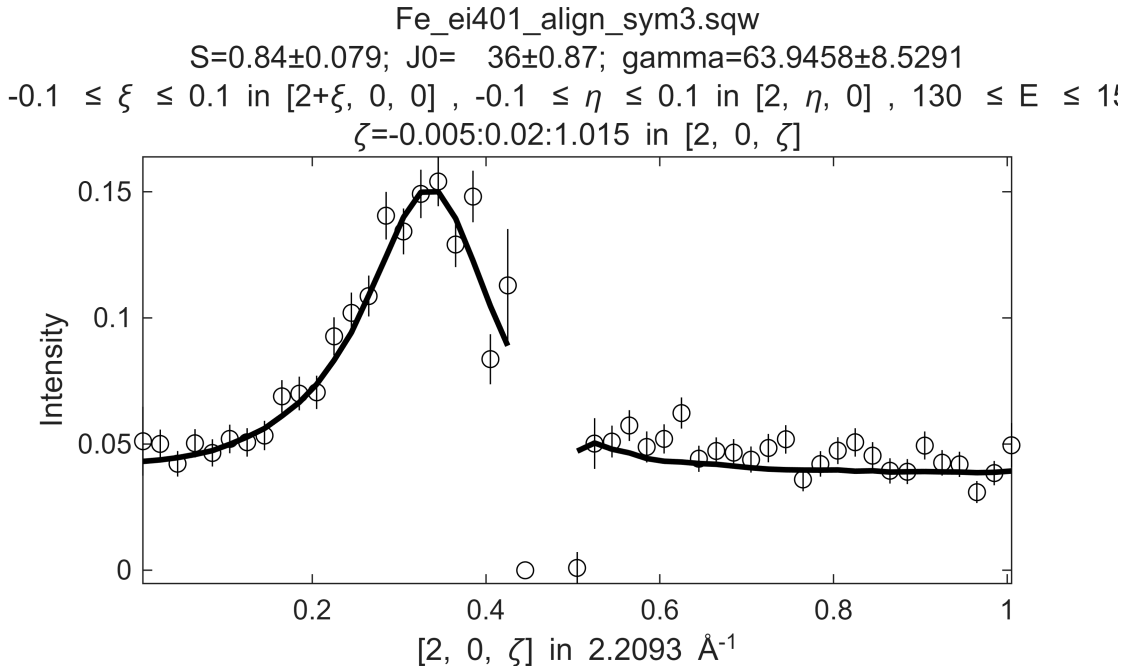
```
*****
**** En = 140±10
*****
Warning: Dataset:3 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 0.98039
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 201122 pixels -- processing data... -----> included 168310 pixels
-----
Number of points in input object: 201122
  Fraction of object selected: 100.00% (= 201122 points) ; Access speed: 2.03GB/sec
  Fraction of object retained: 83.69% (= 168310 points)
```

```

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 34.3%; Transf. time frac.: 62.0%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for  201052 pixels -- processing data... -----> retained  168274 pixels
-----
Number of points in input object: 201052
      Fraction of object selected:  100.00% (=  201052 points) ; Access speed:      1.89GB/sec
      Fraction of object retained:   83.70% (=  168274 points)

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 23.4%; Transf. time frac.: 56.4%; Resulting pix preparation: 20.2%
-----

```



```

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for  201122 pixels -- processing data... -----> retained  168310 pixels
-----
Number of points in input object: 201122
      Fraction of object selected:  100.00% (=  201122 points) ; Access speed:      1.50GB/sec
      Fraction of object retained:   83.69% (=  168310 points)

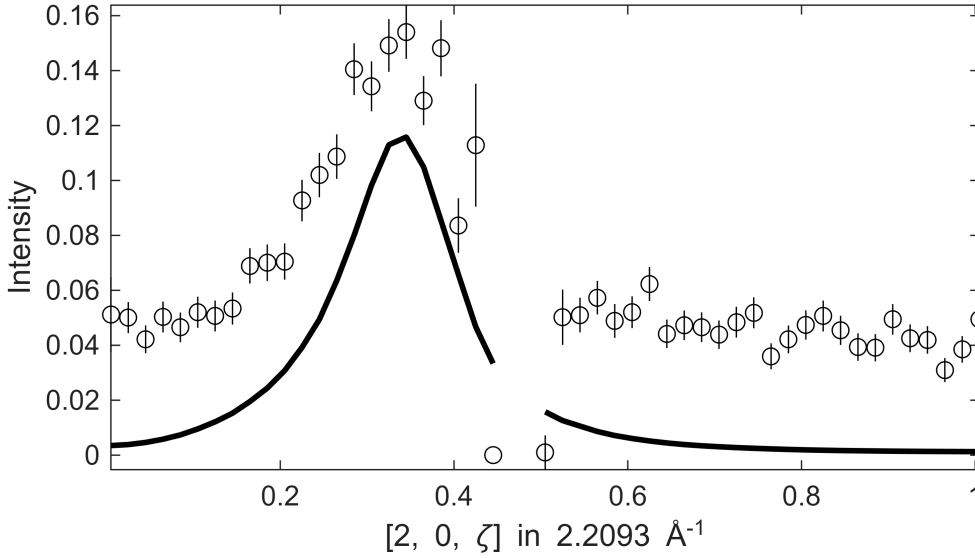
**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 22.3%; Transf. time frac.: 59.7%; Resulting pix preparation: 18.0%
-----

```

Fe_ei401_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $130 \leq E \leq 140$ eV

$\zeta = -0.005:0.02:1.015$ in $[2, 0, \zeta]$



**** En = 150±10

Warning: Dataset:3 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.98039

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 210628 pixels -- processing data... -----> included 174898 pixels

Number of points in input object: 210628

Fraction of object selected: 100.00% (= 210628 points) ; Access speed: 1.63GB/sec

Fraction of object retained: 83.04% (= 174898 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 41.1%; Transf. time frac.: 56.6%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 210417 pixels -- processing data... -----> retained 174862 pixels

Number of points in input object: 210417

Fraction of object selected: 100.00% (= 210417 points) ; Access speed: 1.77GB/sec

Fraction of object retained: 83.10% (= 174862 points)

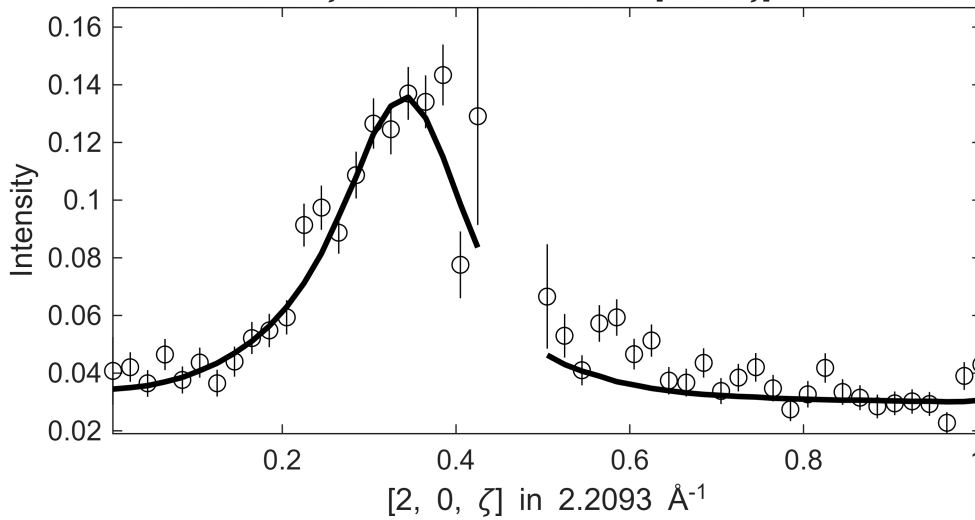
**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 24.7%; Transf. time frac.: 56.3%; Resulting pix preparation: 19.0%

Fe_ei401_align_sym3.sqw

S=0.88±0.031; J0= 38±0.78; gamma=72.1153±1.82154

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 140 ≤ E ≤ 160
 ζ=-0.005:0.02:1.015 in [2, 0, ζ]

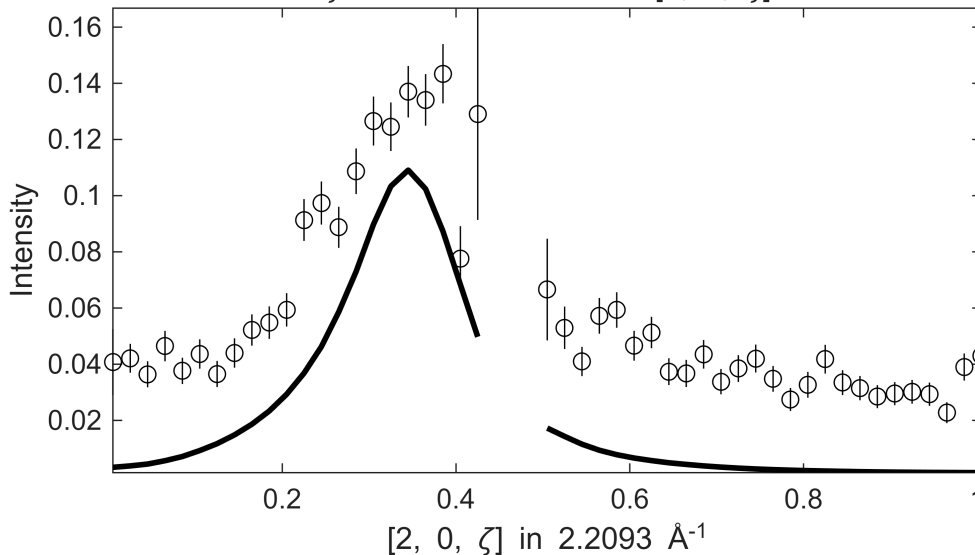


```
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 210628 pixels -- processing data... -----> retained 174898 pixels
-----
Number of points in input object: 210628
  Fraction of object selected: 100.00% (= 210628 points) ; Access speed: 1.66GB/sec
  Fraction of object retained: 83.04% (= 174898 points)

**** Total time in cut(sqw) : 0.0sec Including:
  Data access time fraction: 27.4%; Transf. time frac.: 53.0%; Resulting pix preparation: 19.6%
-----
```

Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 140 ≤ E ≤ 160
 ζ=-0.005:0.02:1.015 in [2, 0, ζ]



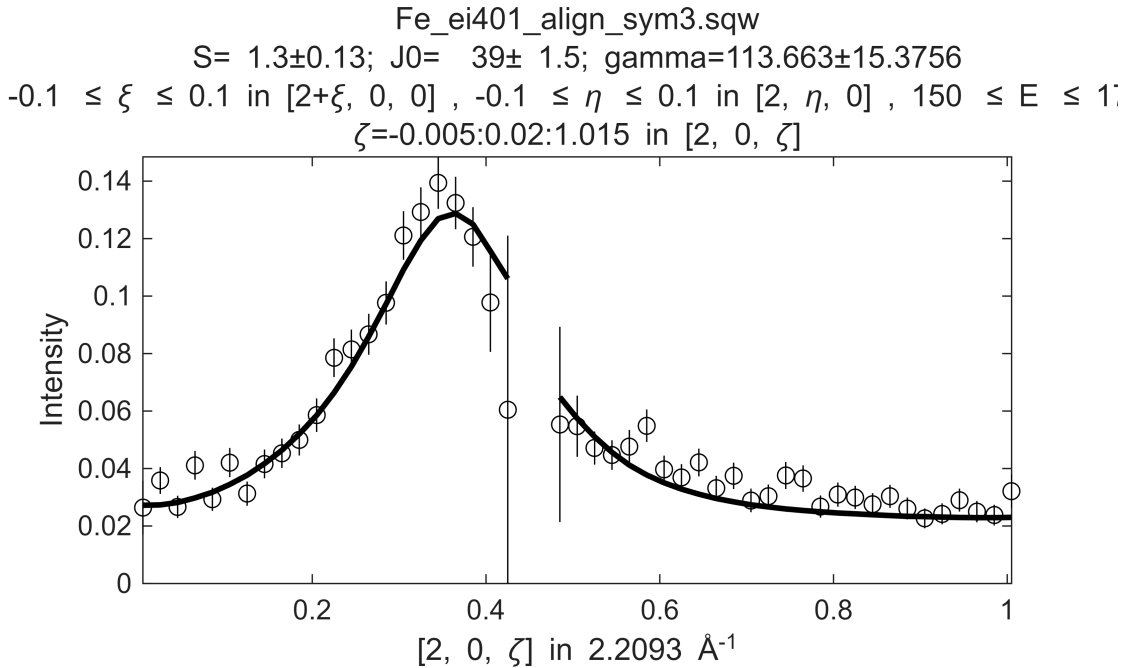
```
*****
**** En = 160±10
*****
Warning: Dataset:5 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 1.634
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 225504 pixels -- processing data... -----> included 187296 pixels
-----
Number of points in input object: 225504
  Fraction of object selected: 100.00% (= 225504 points) ; Access speed: 1.93GB/sec
  Fraction of object retained: 83.06% (= 187296 points)
```

```

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 39.3%; Transf. time frac.: 57.9%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for  225062 pixels -- processing data... -----> retained  187055 pixels
-----
Number of points in input object: 225062
      Fraction of object selected:  100.00% (=  225062 points) ; Access speed:      1.88GB/sec
      Fraction of object retained:   83.11% (=  187055 points)

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 23.8%; Transf. time frac.: 58.6%; Resulting pix preparation: 17.5%
-----

```



```

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for  225504 pixels -- processing data... -----> retained  187296 pixels
-----
Number of points in input object: 225504
      Fraction of object selected:  100.00% (=  225504 points) ; Access speed:      1.64GB/sec
      Fraction of object retained:   83.06% (=  187296 points)

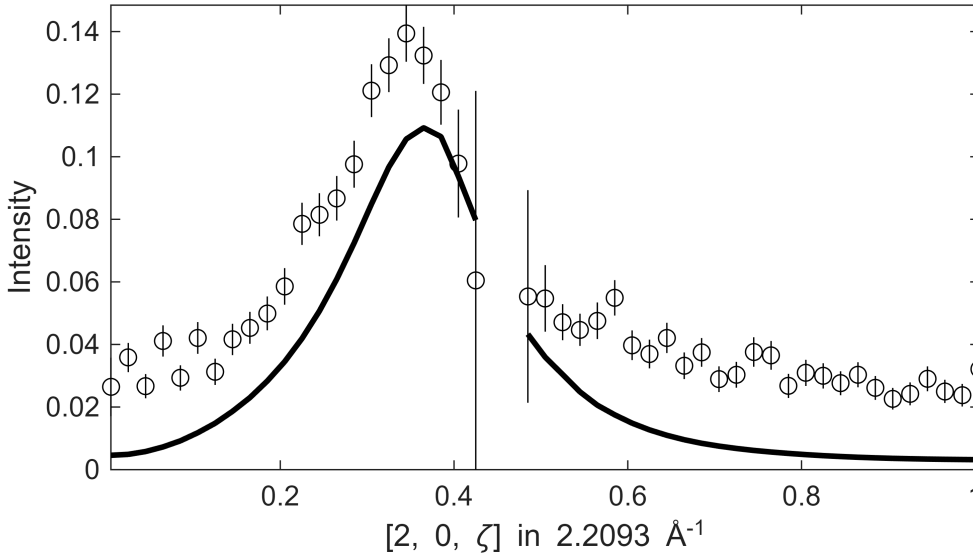
**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 28.9%; Transf. time frac.: 50.7%; Resulting pix preparation: 20.4%
-----

```


Fe_ei401_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $150 \leq E \leq 170$ keV

$\zeta = -0.005:0.02:1.015$ in $[2, 0, \zeta]$



**** En = 170±10

Warning: Dataset:4 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.3072

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 254158 pixels -- processing data... -----> included 210442 pixels

Number of points in input object: 254158

Fraction of object selected: 100.00% (= 254158 points) ; Access speed: 1.38GB/sec

Fraction of object retained: 82.80% (= 210442 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 44.4%; Transf. time frac.: 52.6%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 253623 pixels -- processing data... -----> retained 209999 pixels

Number of points in input object: 253623

Fraction of object selected: 100.00% (= 253623 points) ; Access speed: 1.93GB/sec

Fraction of object retained: 82.80% (= 209999 points)

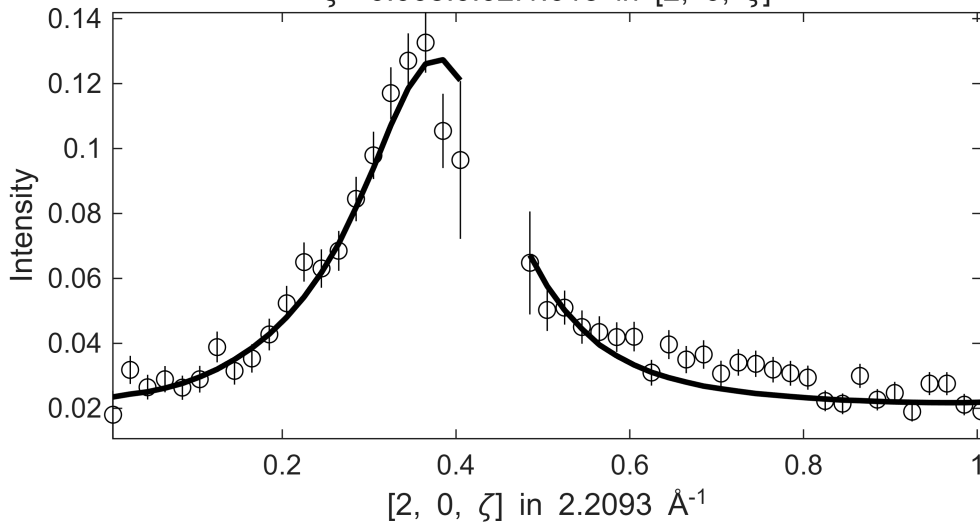
**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 23.7%; Transf. time frac.: 56.8%; Resulting pix preparation: 19.5%

Fe_ei401_align_sym3.sqw

S= 1.1±0.045; J0= 37±0.55; gamma=90.9796±2.55759

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 160 ≤ E ≤ 180
ζ=-0.005:0.02:1.015 in [2, 0, ζ]

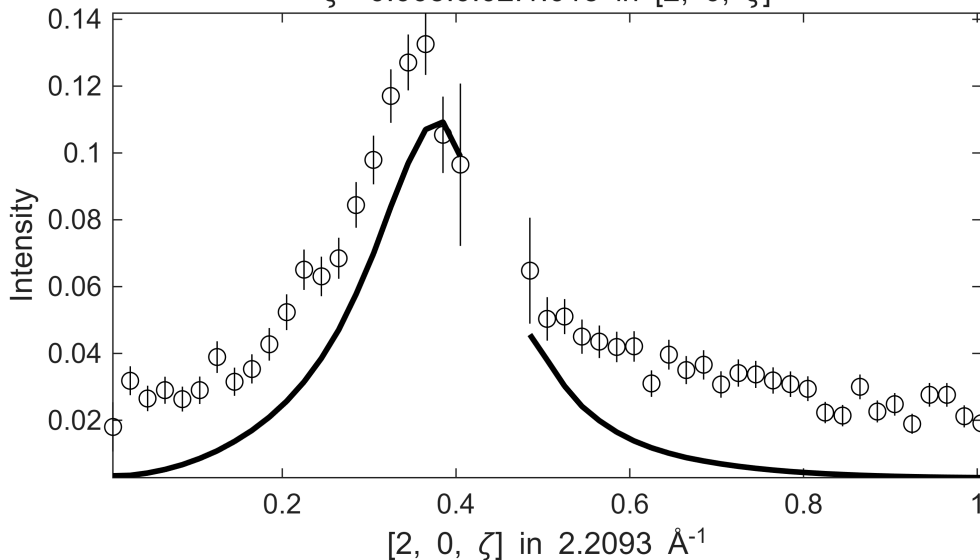


```
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 254158 pixels -- processing data... -----> retained 210442 pixels
-----
Number of points in input object: 254158
  Fraction of object selected: 100.00% (= 254158 points) ; Access speed: 1.63GB/sec
  Fraction of object retained: 82.80% (= 210442 points)

**** Total time in cut(sqw) : 0.0sec Including:
  Data access time fraction: 24.4%; Transf. time frac.: 52.6%; Resulting pix preparation: 22.9%
-----
```

Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 160 ≤ E ≤ 180
ζ=-0.005:0.02:1.015 in [2, 0, ζ]



```
*****
**** En = 180±10
*****
Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 0.65359
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 310312 pixels -- processing data... -----> included 256038 pixels
-----
Number of points in input object: 310312
  Fraction of object selected: 100.00% (= 310312 points) ; Access speed: 1.94GB/sec
  Fraction of object retained: 82.51% (= 256038 points)
```

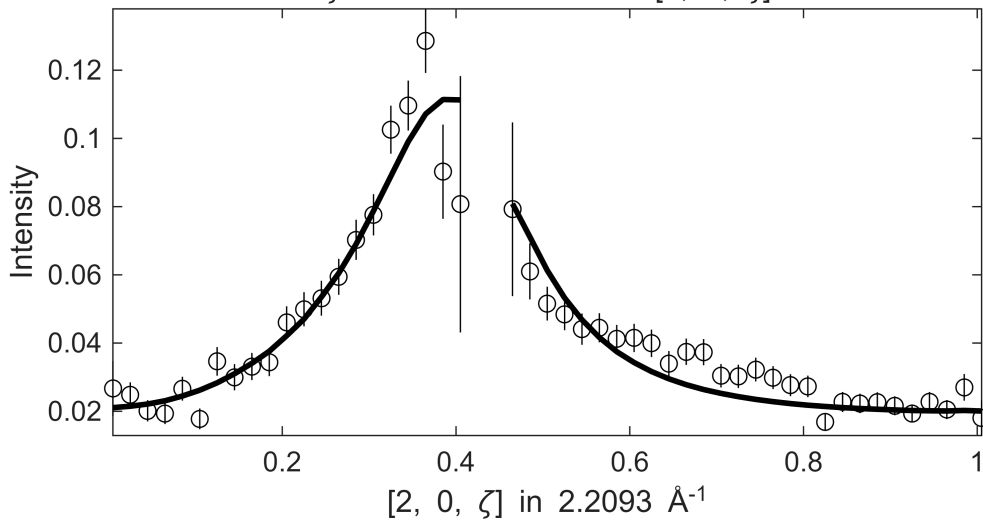
```

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 42.7%; Transf. time frac.: 54.5%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for  310134 pixels -- processing data... -----> retained  255860 pixels
-----
Number of points in input object: 310134
      Fraction of object selected:  100.00% (=  310134 points) ; Access speed:      1.95GB/sec
      Fraction of object retained:   82.50% (=  255860 points)

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 23.1%; Transf. time frac.: 59.5%; Resulting pix preparation: 17.4%
-----

```

Fe_ei401_align_sym3.sqw
 $S = 1.1 \pm 0.087$; $J_0 = 37 \pm 1.3$; $\gamma = 107.381 \pm 14.5762$
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $170 \leq E \leq 190$ eV
 $\zeta = -0.005:0.02:1.015$ in $[2, 0, \zeta]$



```

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for  310312 pixels -- processing data... -----> retained  256038 pixels
-----
Number of points in input object: 310312
      Fraction of object selected:  100.00% (=  310312 points) ; Access speed:      1.97GB/sec
      Fraction of object retained:   82.51% (=  256038 points)

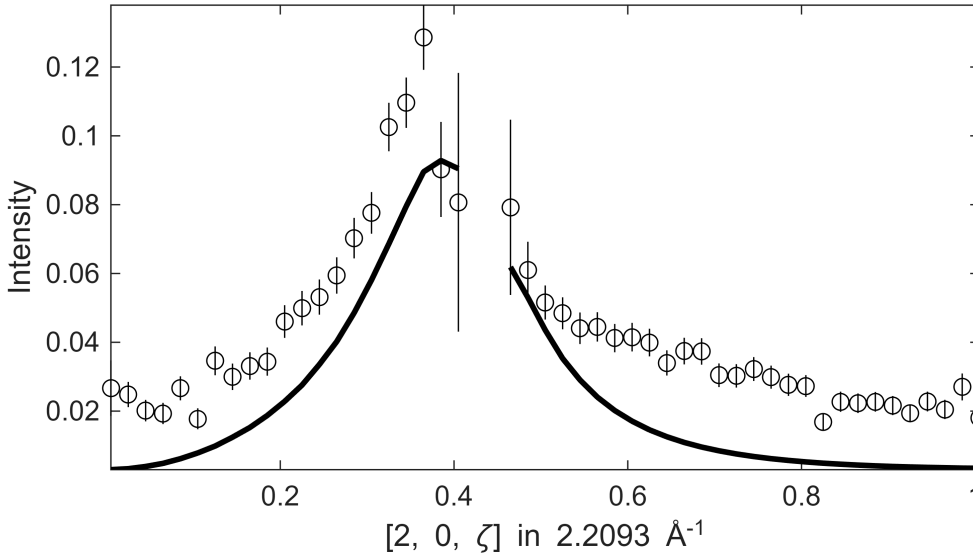
**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 22.9%; Transf. time frac.: 60.1%; Resulting pix preparation: 17.0%
-----

```

Fe_ei401_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $170 \leq E \leq 190$

$\zeta = -0.005:0.02:1.015$ in $[2, 0, \zeta]$



**** En = 190±10

Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.65359

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 420351 pixels -- processing data... -----> included 340426 pixels

Number of points in input object: 420351

Fraction of object selected: 100.00% (= 420351 points) ; Access speed: 2.00GB/sec

Fraction of object retained: 80.99% (= 340426 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 43.8%; Transf. time frac.: 54.2%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 420162 pixels -- processing data... -----> retained 340289 pixels

Number of points in input object: 420162

Fraction of object selected: 100.00% (= 420162 points) ; Access speed: 2.01GB/sec

Fraction of object retained: 80.99% (= 340289 points)

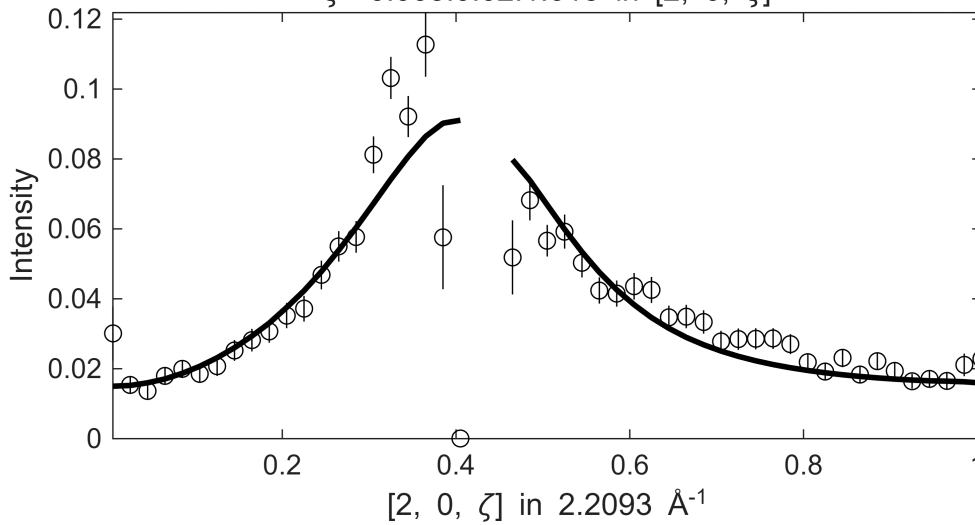
**** Total time in cut(sqw) : 0.1sec Including:

Data access time fraction: 25.6%; Transf. time frac.: 57.3%; Resulting pix preparation: 17.1%

Fe_ei401_align_sym3.sqw

S= 1.7±0.25; J0= 43± 3.5; gamma=206.947±47.7933

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 180 ≤ E ≤ 200
ζ=-0.005:0.02:1.015 in [2, 0, ζ]

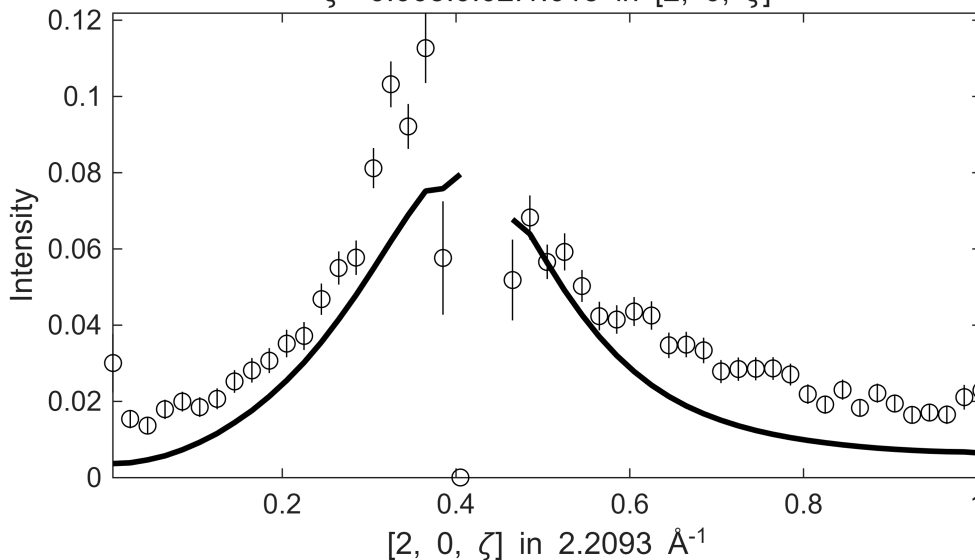


```
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 420351 pixels -- processing data... -----> retained 340426 pixels
-----
Number of points in input object: 420351
  Fraction of object selected: 100.00% (= 420351 points) ; Access speed: 0.62GB/sec
  Fraction of object retained: 80.99% (= 340426 points)

**** Total time in cut(sqw) : 0.1sec Including:
  Data access time fraction: 42.2%; Transf. time frac.: 37.8%; Resulting pix preparation: 19.9%
-----
```

Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 180 ≤ E ≤ 200
ζ=-0.005:0.02:1.015 in [2, 0, ζ]



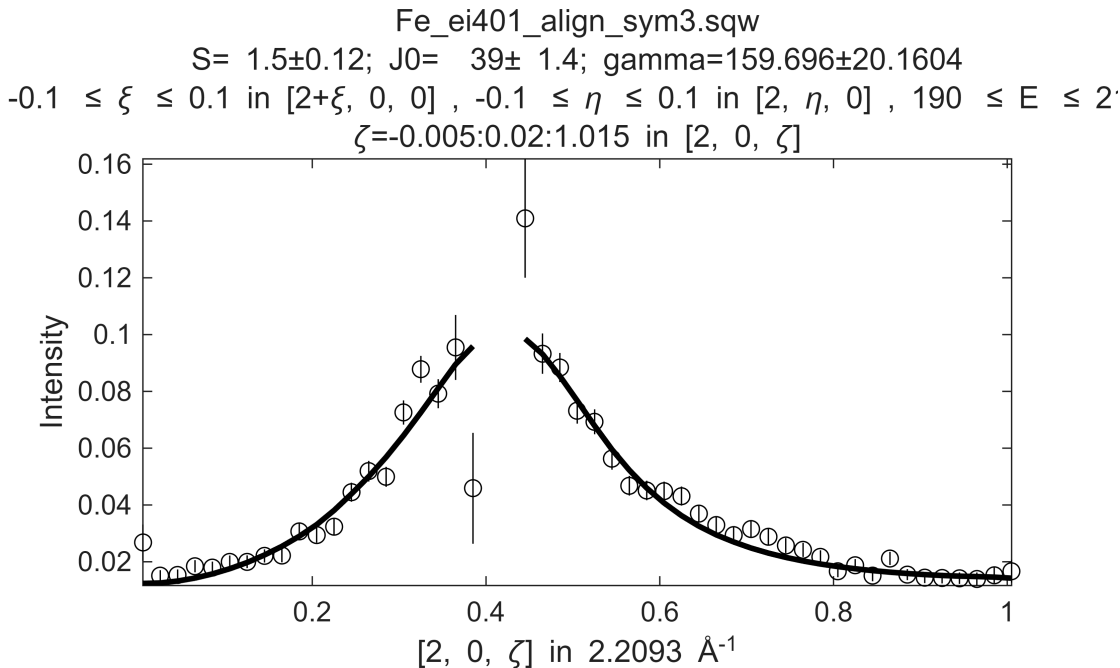
```
*****
**** En = 200±10
*****
Warning: Dataset:3 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 0.98039
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 585849 pixels -- processing data... -----> included 500457 pixels
-----
Number of points in input object: 585849
  Fraction of object selected: 100.00% (= 585849 points) ; Access speed: 2.05GB/sec
  Fraction of object retained: 85.42% (= 500457 points)
```

```

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction:  47.1%; Transf. time frac.:  51.3%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for  585121 pixels -- processing data... -----> retained  499925 pixels
-----
Number of points in input object: 585121
      Fraction of object selected:  100.00% (=  585121 points) ; Access speed:      2.11GB/sec
      Fraction of object retained:   85.44% (=  499925 points)

**** Total time in cut(sqw)   :      0.1sec Including:
      Data access time fraction:  27.2%; Transf. time frac.:  53.3%; Resulting pix preparation:  19.4%
-----

```



```

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for  585849 pixels -- processing data... -----> retained  500457 pixels
-----
Number of points in input object: 585849
      Fraction of object selected:  100.00% (=  585849 points) ; Access speed:      2.10GB/sec
      Fraction of object retained:   85.42% (=  500457 points)

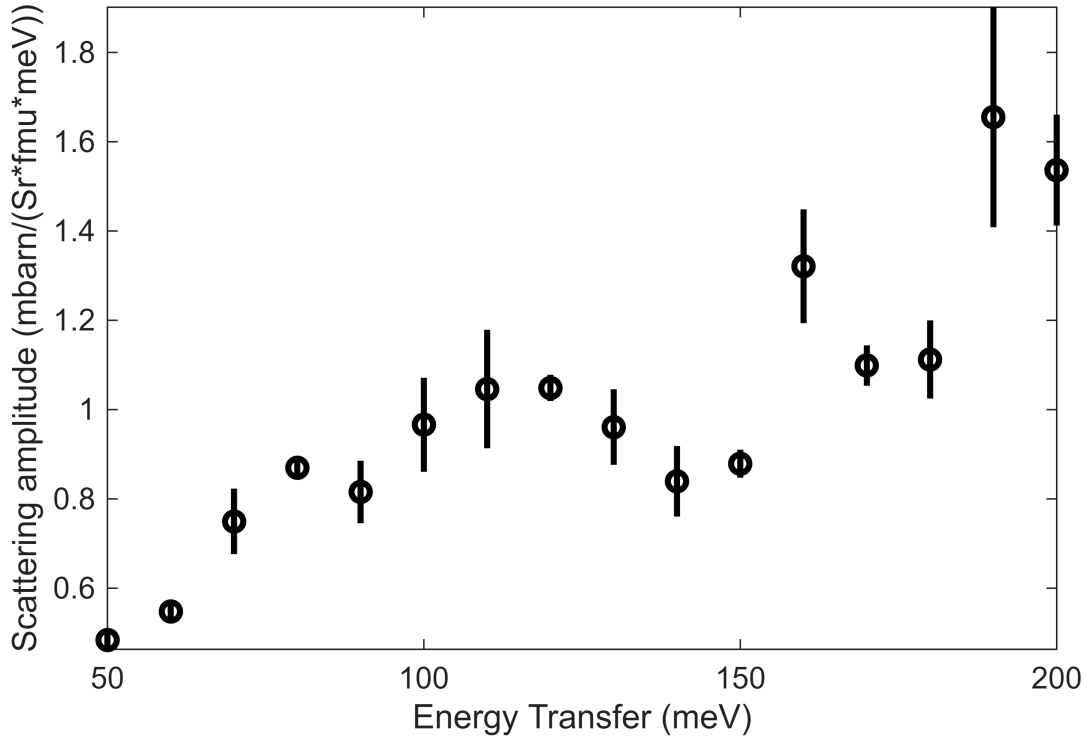
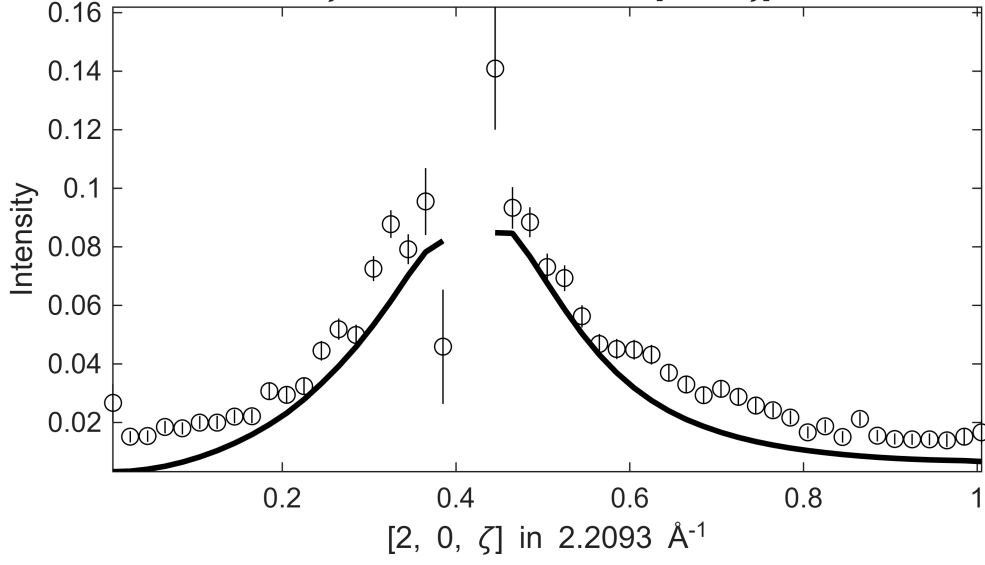
**** Total time in cut(sqw)   :      0.1sec Including:
      Data access time fraction:  26.5%; Transf. time frac.:  51.7%; Resulting pix preparation:  21.8%
-----

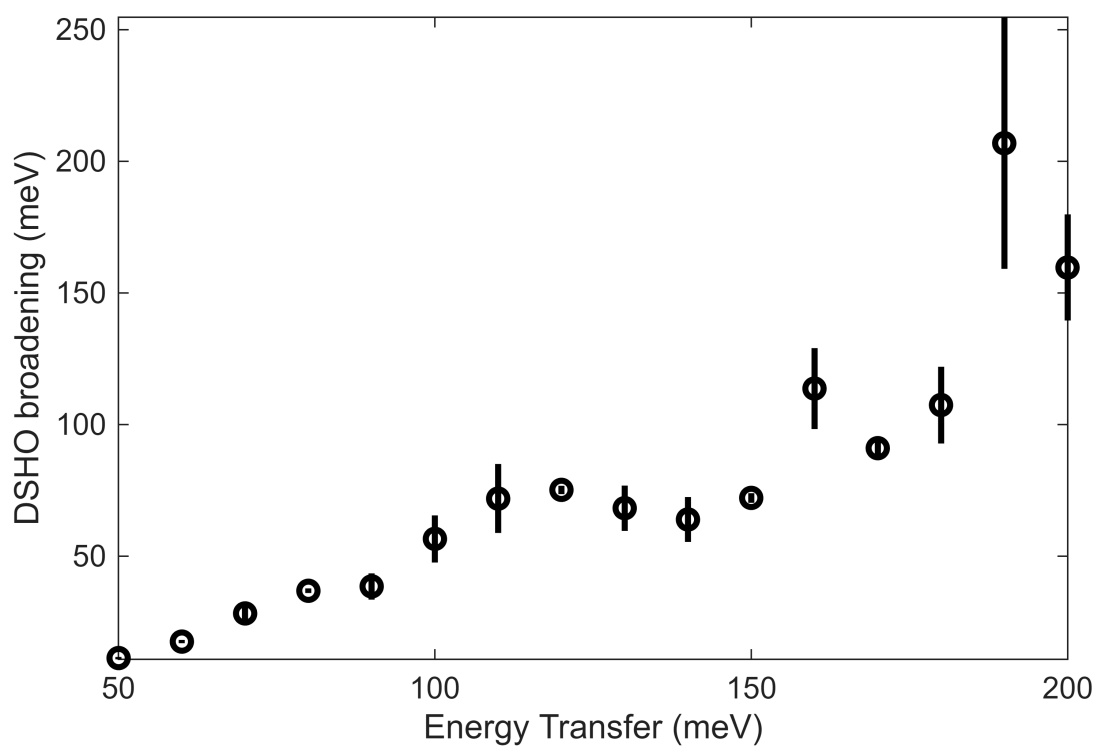
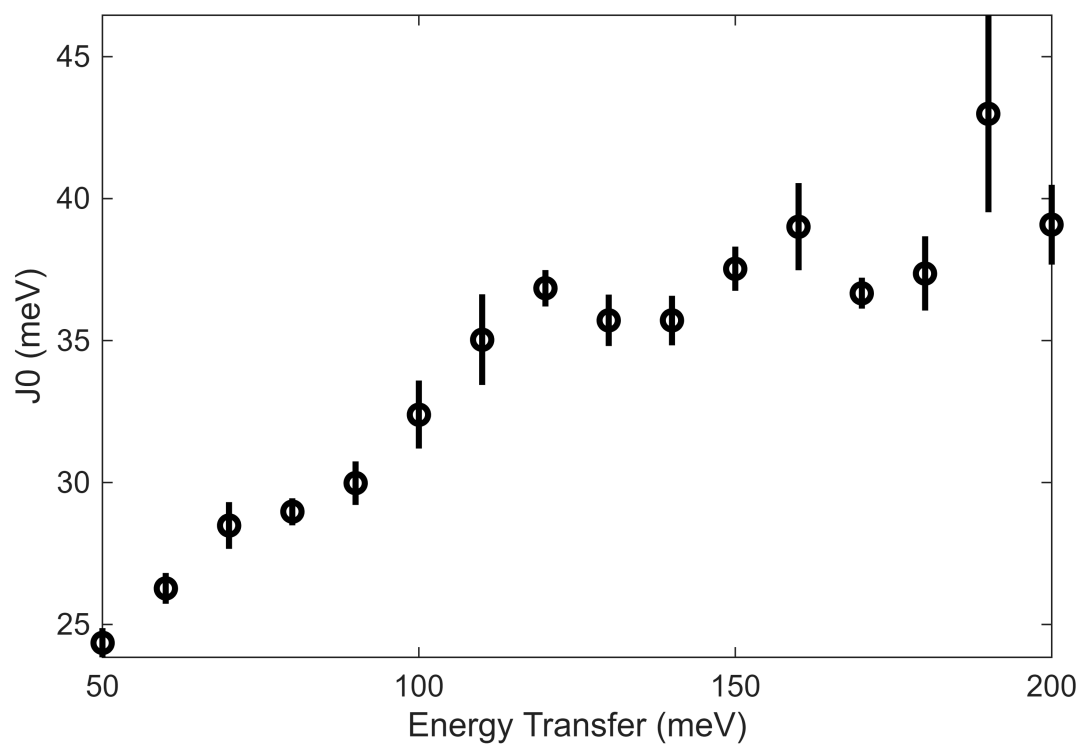
```

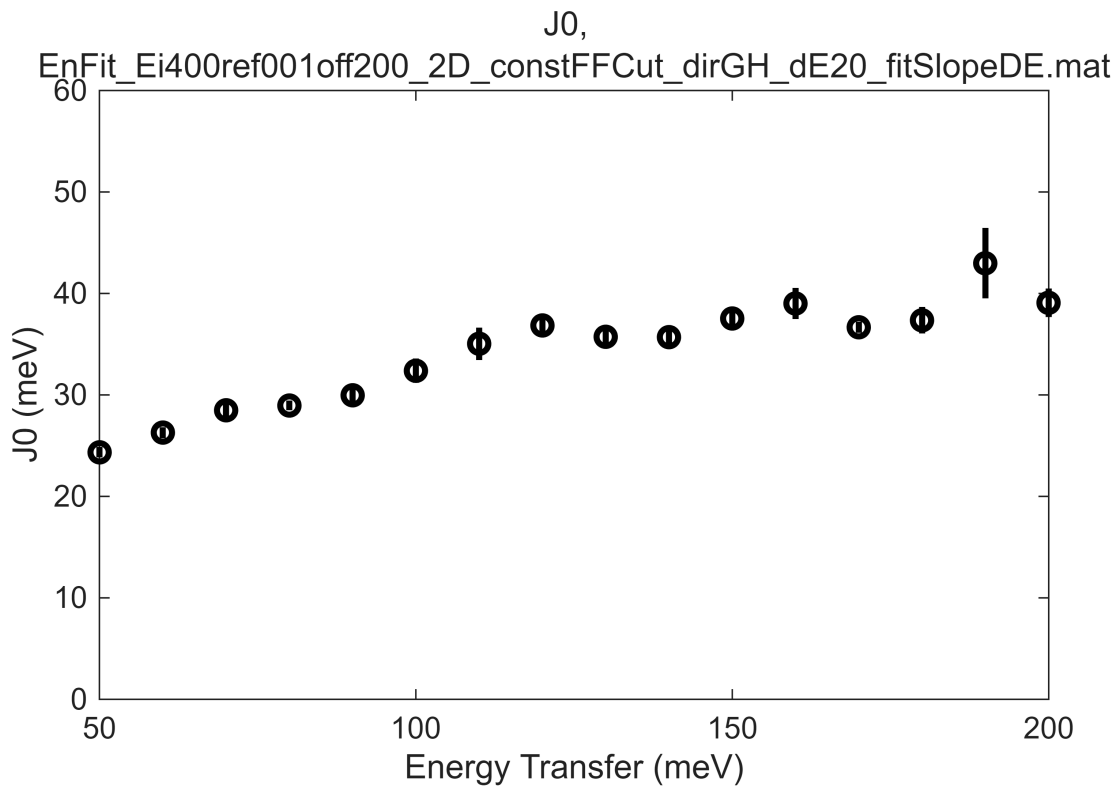
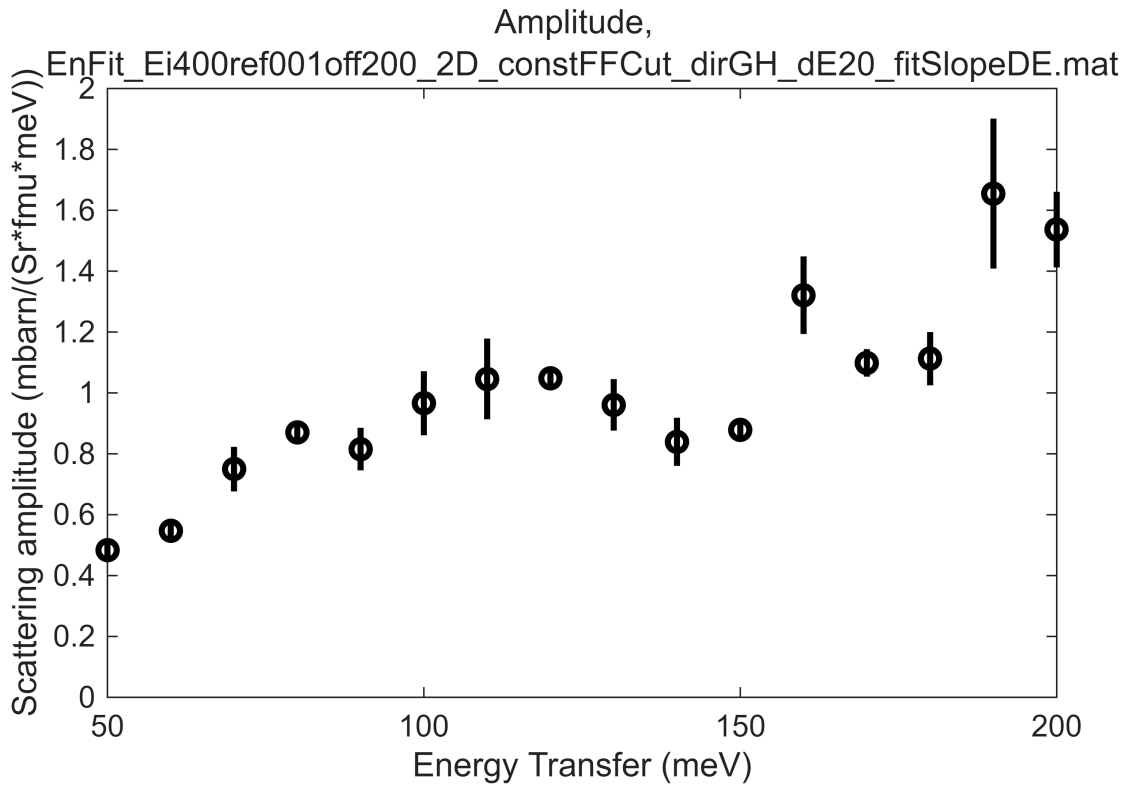
Fe_ei401_align_sym3.sqw

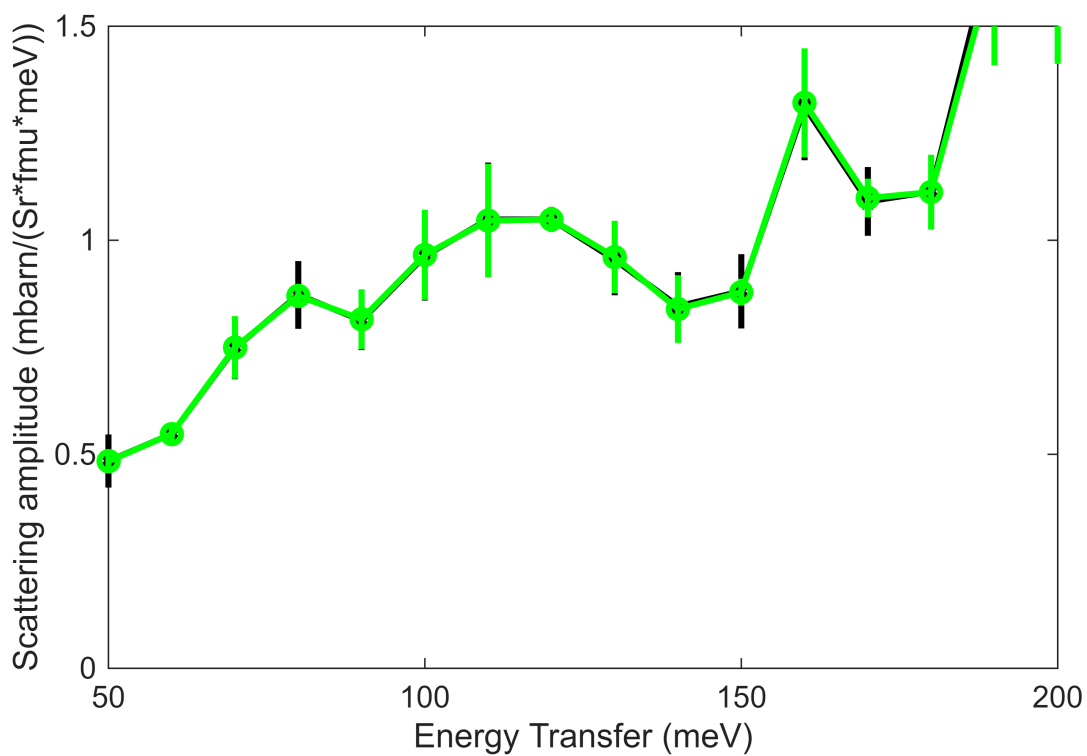
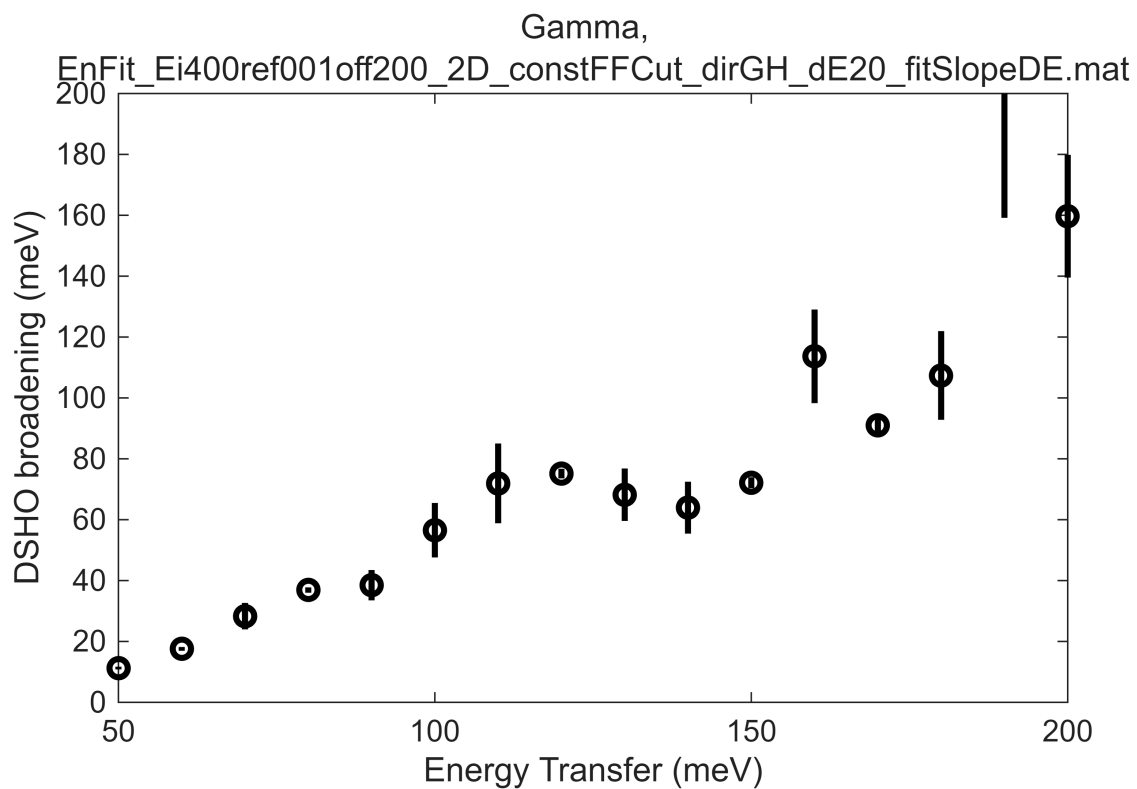
$-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $190 \leq E \leq 210$ meV

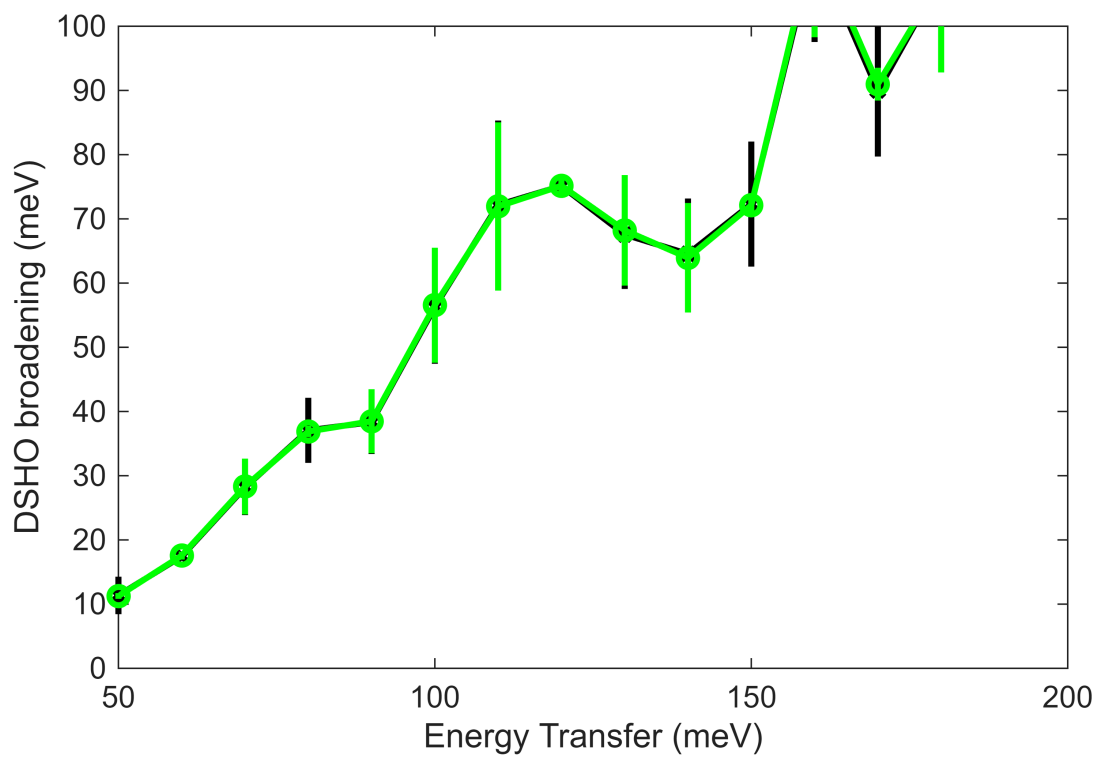
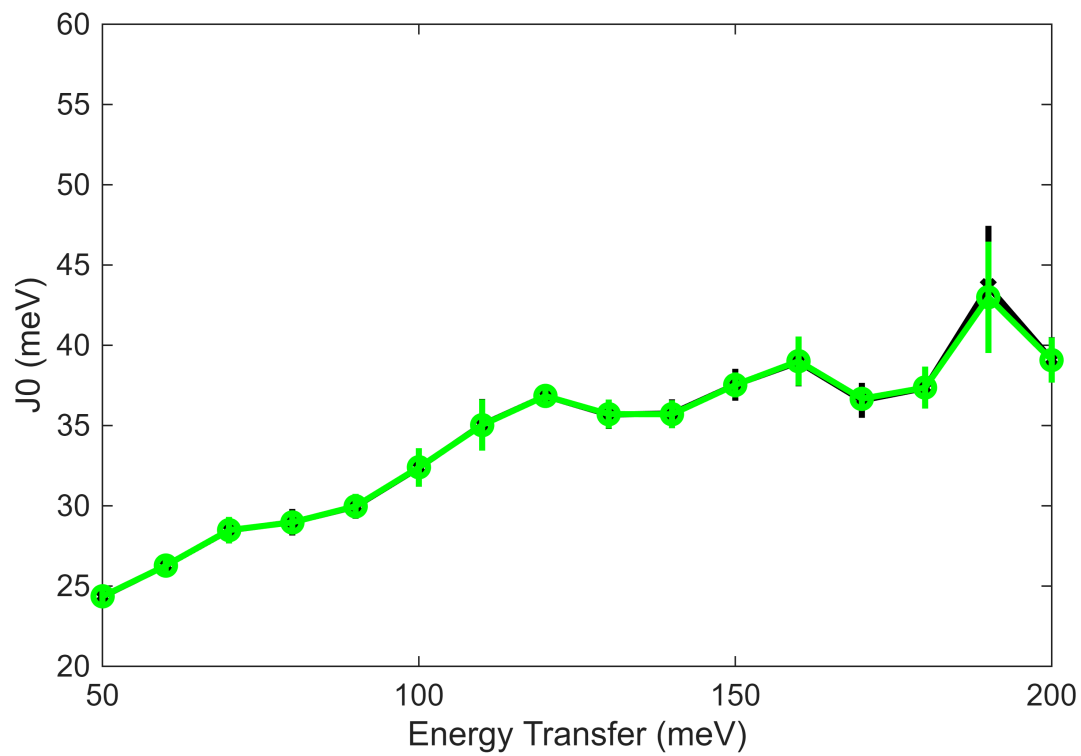
$\zeta = -0.005:0.02:1.015$ in $[2, 0, \zeta]$











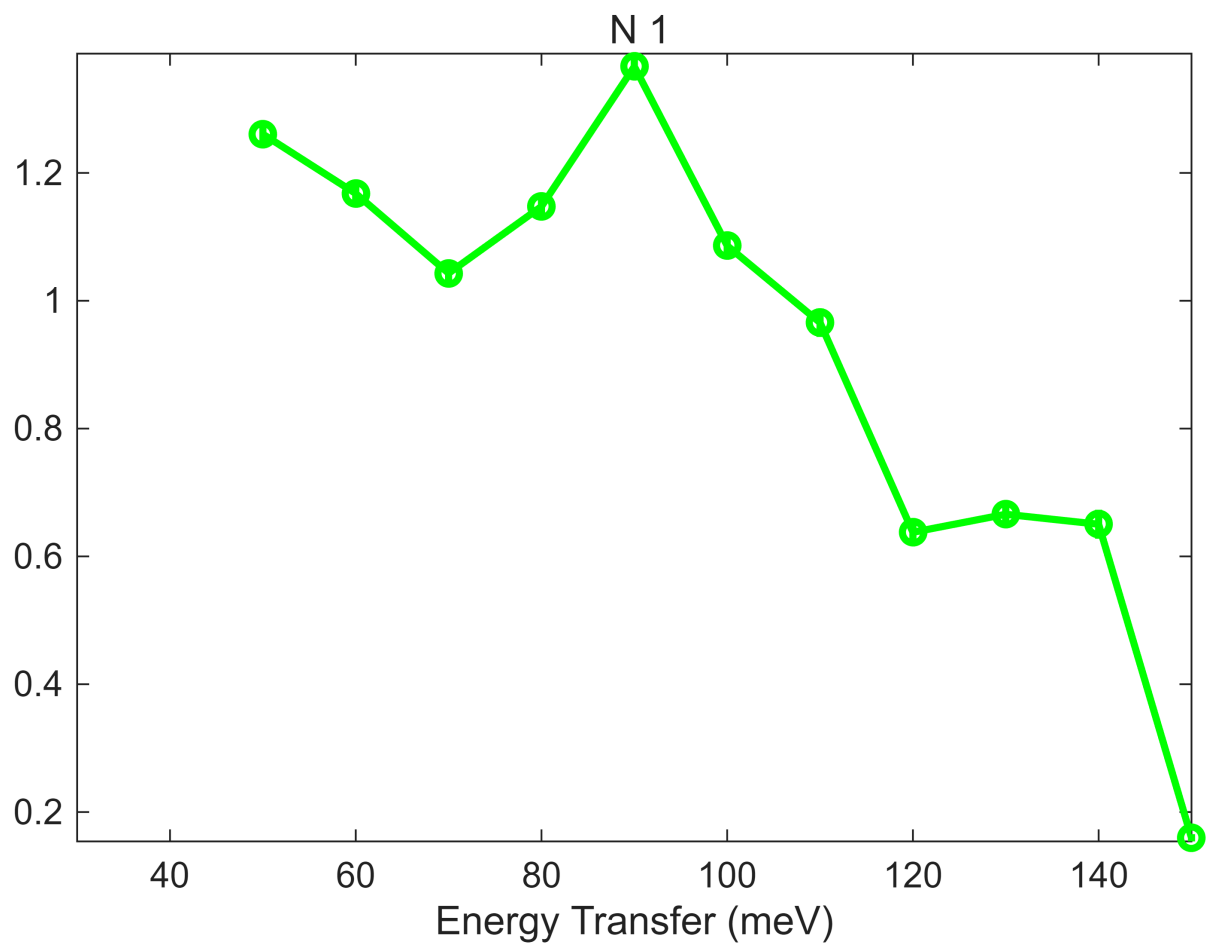
```
%c1,'En_cuts800',dir_name,cut_en,dE_step,half_dE);
beep
beep
beep
```

```
fit_res_400 = fit_res_400rec;

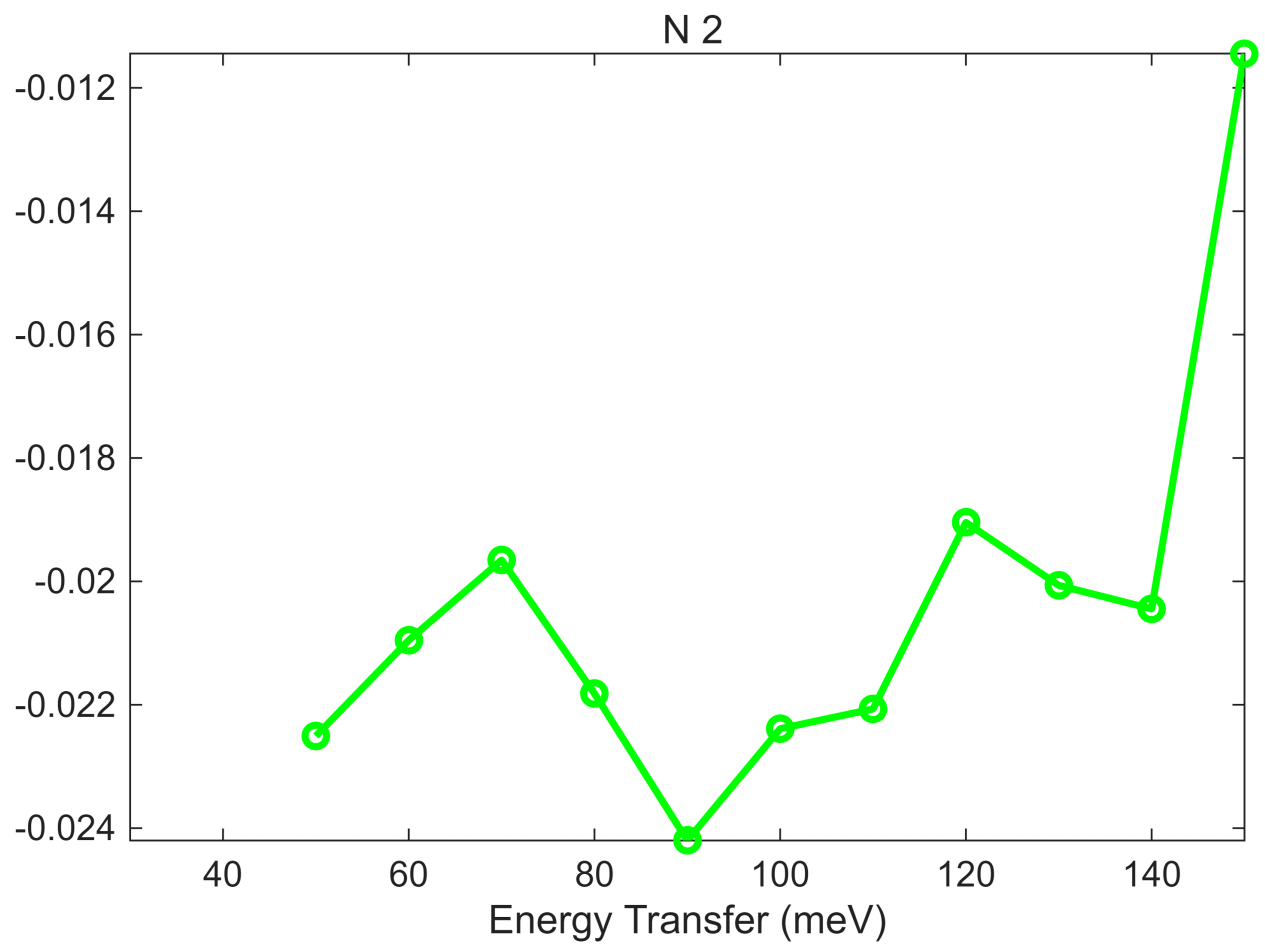
bg_data = extract_bg_par(fit_res_400.all_fit_par,[1,2]);
```

```
% original slope parameters
```

```
pd(bg_data(1));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_f  
figure;
```



```
pd(bg_data(2));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_f  
figure
```



```
fit_resEi400 = fit_res_400rec;  
save(res_name, "fit_resEi400")
```